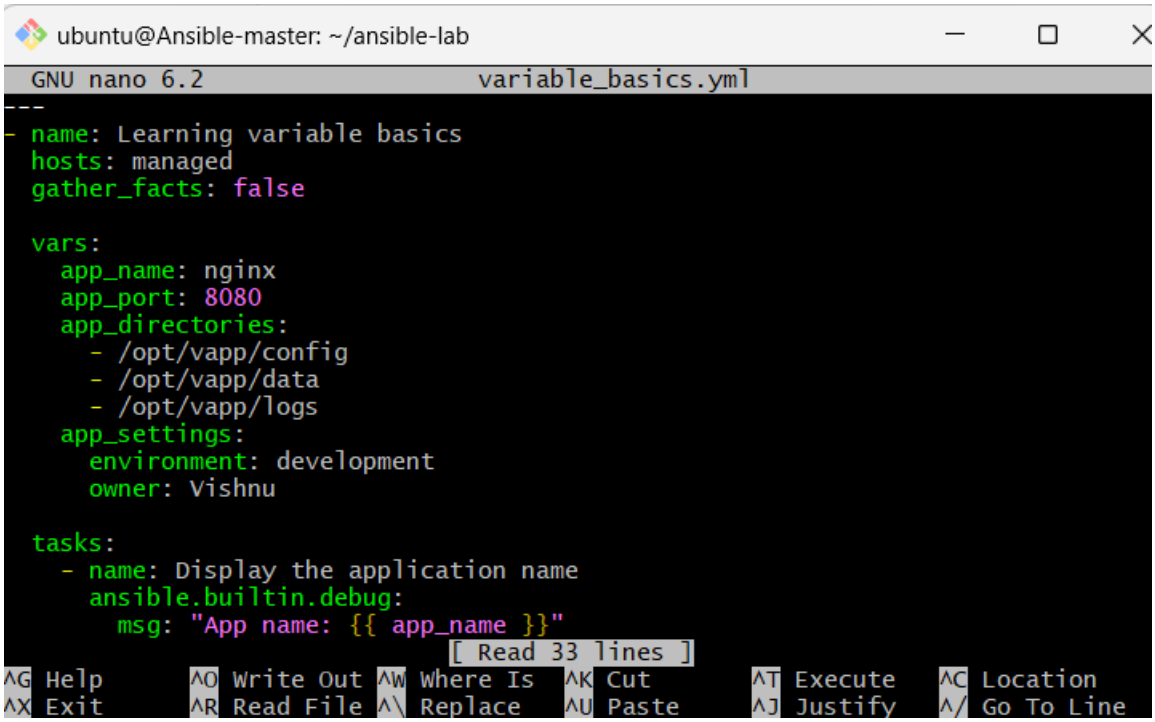
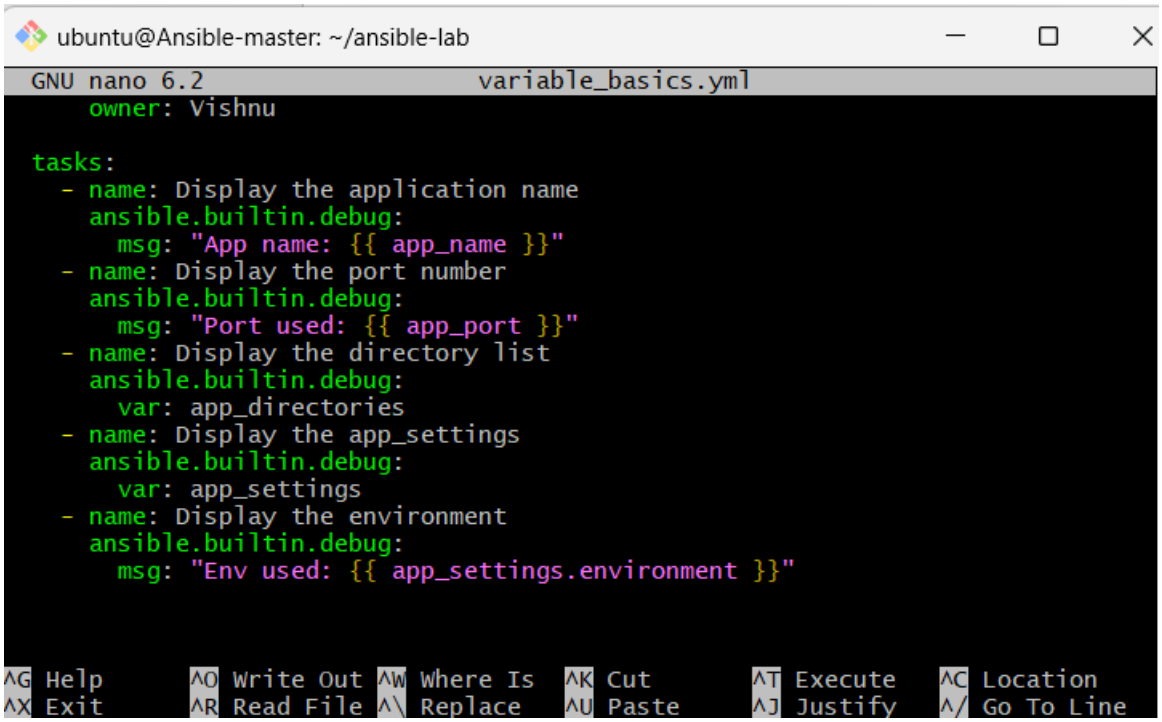
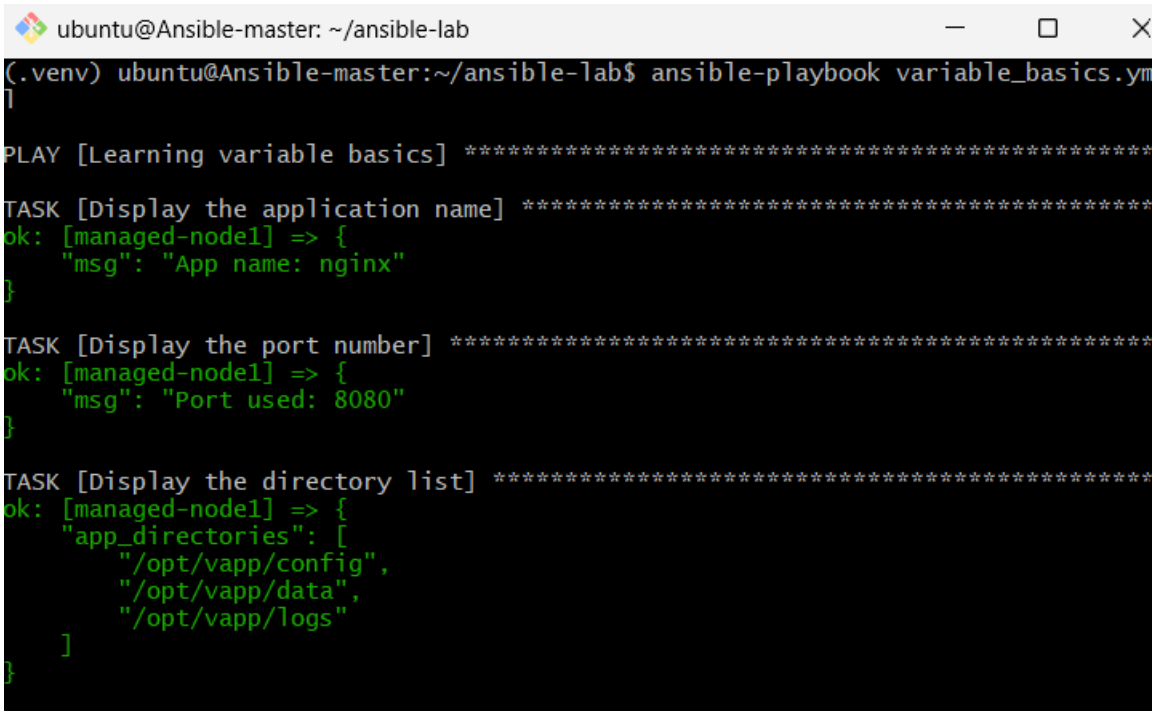
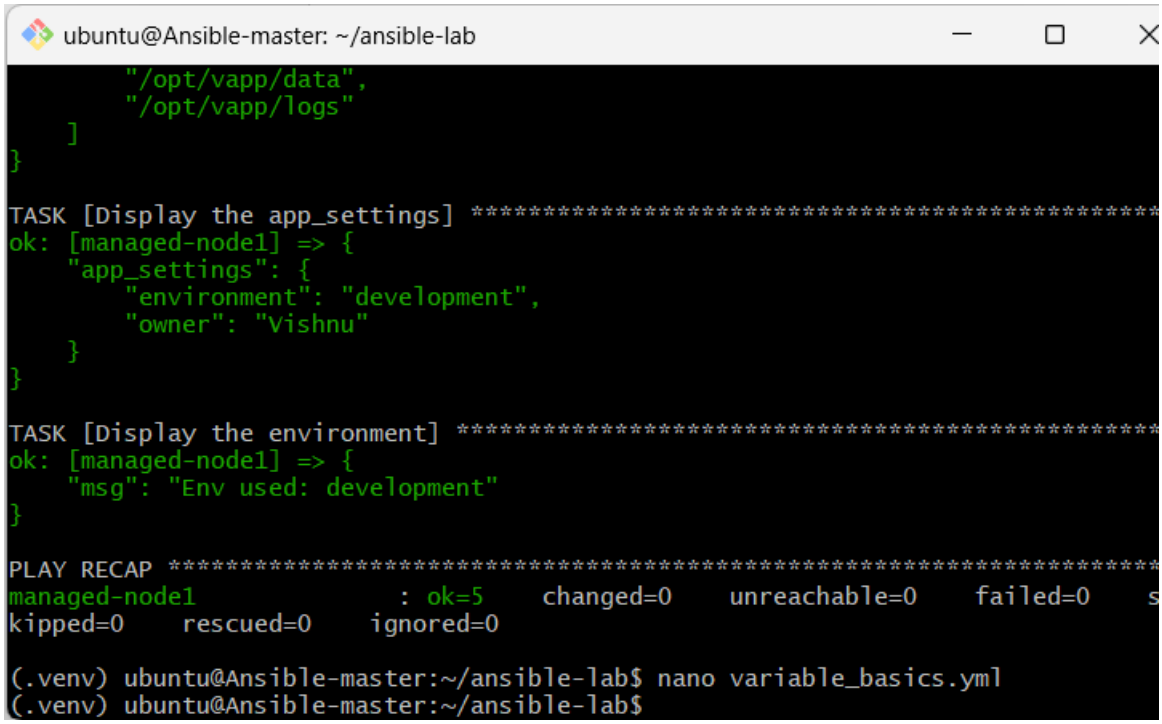
Ansible Mastery Part-4:

Advanced Playbook Development with group_vars and host_vars, System Fact Gathering, Registered Variables, Conditional Task Execution, List and Dictionary Iteration, Idempotent Directory Management, and Dynamic Configuration Generation Using Jinja2 Templates

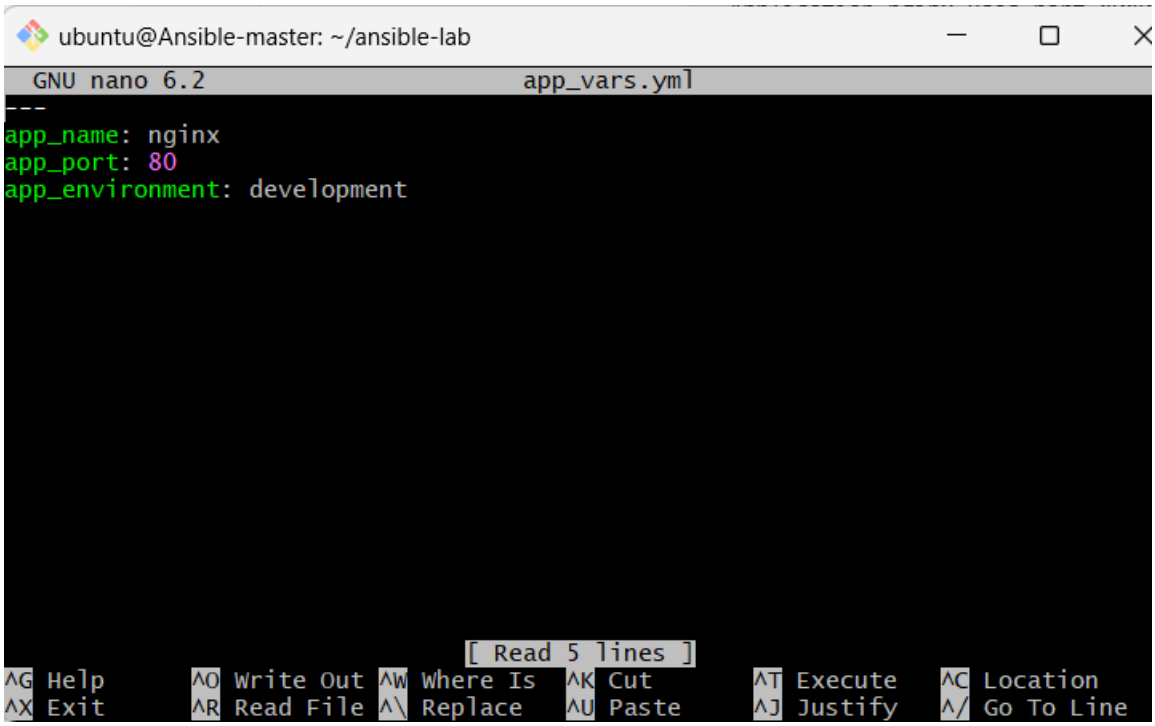


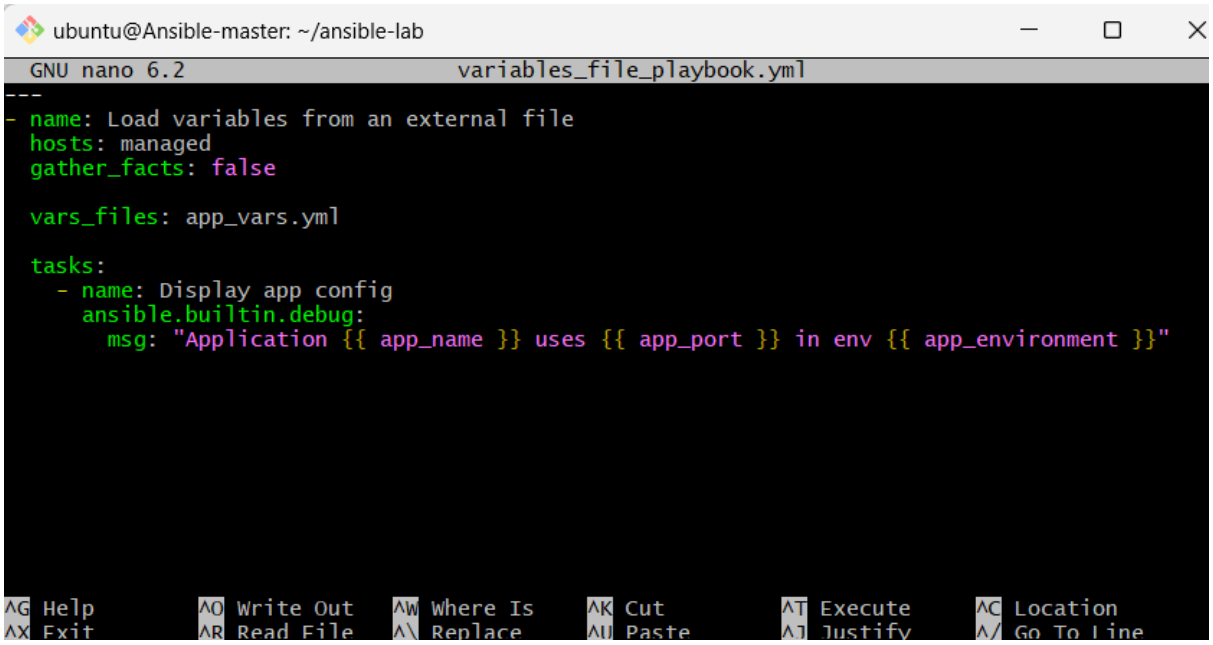

ubuntu@Ansible-master: ~/ansible-lab
GNU nano 6.2 variable_basics.yml
owner: Vishnu
tasks:
- name: Display the application name
ansible.builtin.debug:
msg: "App name: {{ app_name }}"
- name: Display the port number
ansible.builtin.debug:
msg: "Port used: {{ app_port }}"
- name: Display the directory list
ansible.builtin.debug:
var: app_directories
- name: Display the app_settings
ansible.builtin.debug:
var: app_settings
- name: Display the environment
ansible.builtin.debug:
msg: "Env used: {{ app_settings.environment }}"
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line


ubuntu@Ansible-master: ~/ansible-lab
(.venv) ubuntu@Ansible-master:~/ansible-lab$ ansible-playbook variable_basics.ym
l
PLAY [Learning variable basics] ****
TASK [Display the application name] ****
ok: [managed-node1] => {
"msg": "App name: nginx"
}
TASK [Display the port number] ****
ok: [managed-node1] => {
"msg": "Port used: 8080"
}
TASK [Display the directory list] ****
ok: [managed-node1] => {
"app_directories": [
"/opt/vapp/config",
"/opt/vapp/data",
"/opt/vapp/logs"
]
}

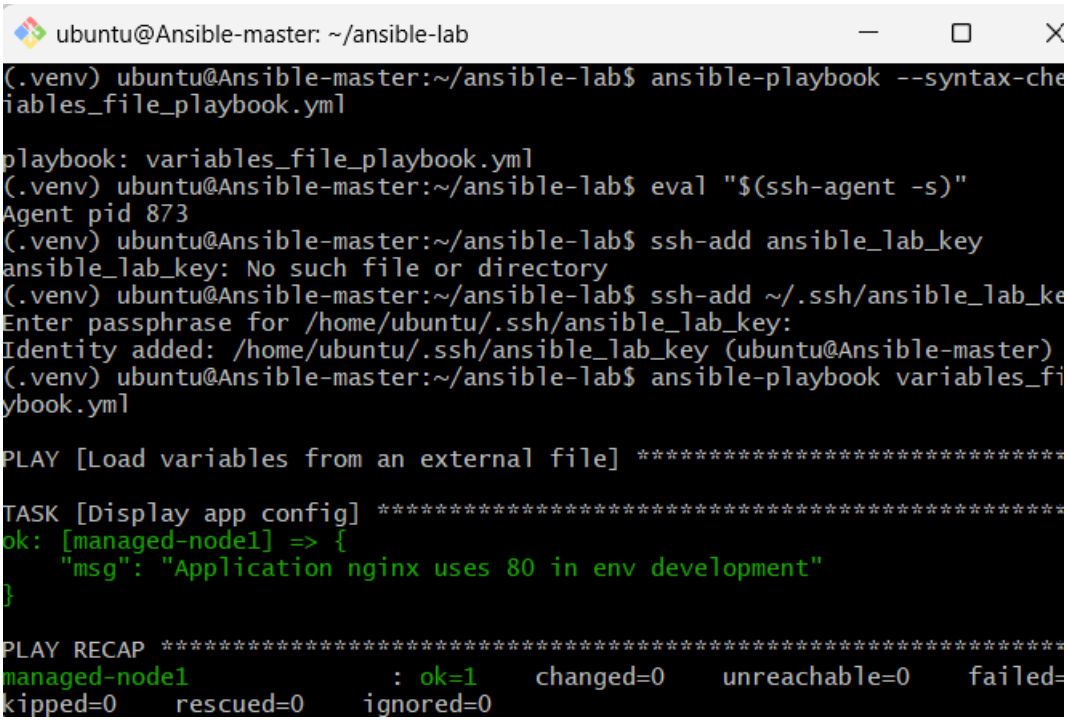


Moving variables into an External file:

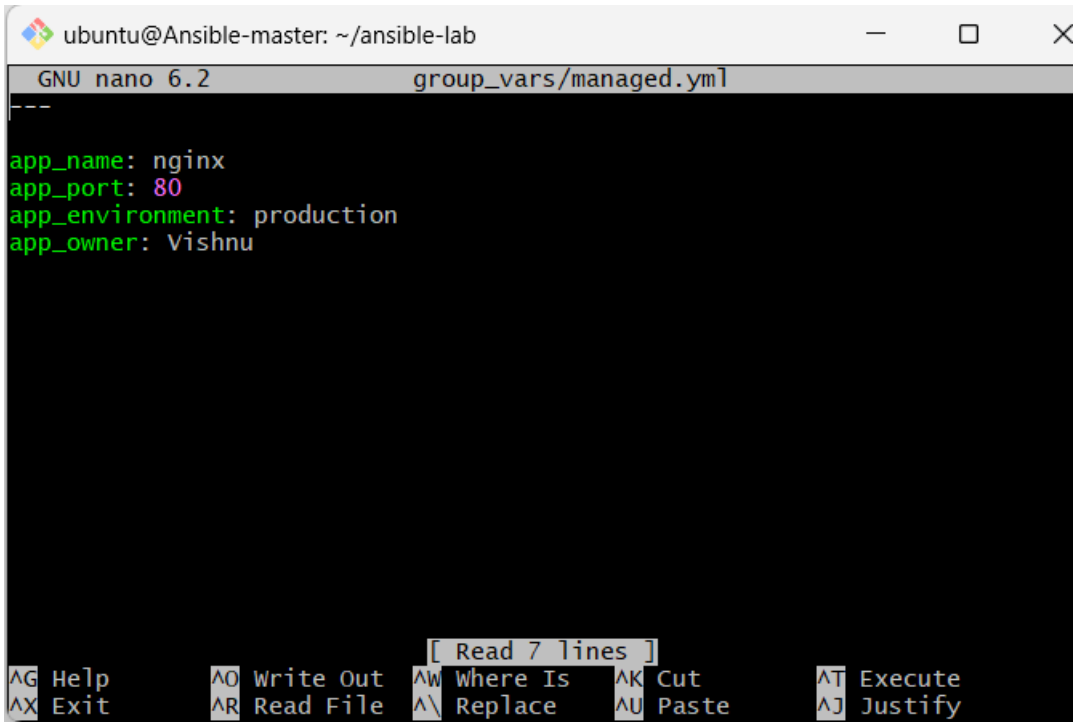



ubuntu@Ansible-master: ~/ansible-lab
GNU nano 6.2 variables_file_playbook.yml

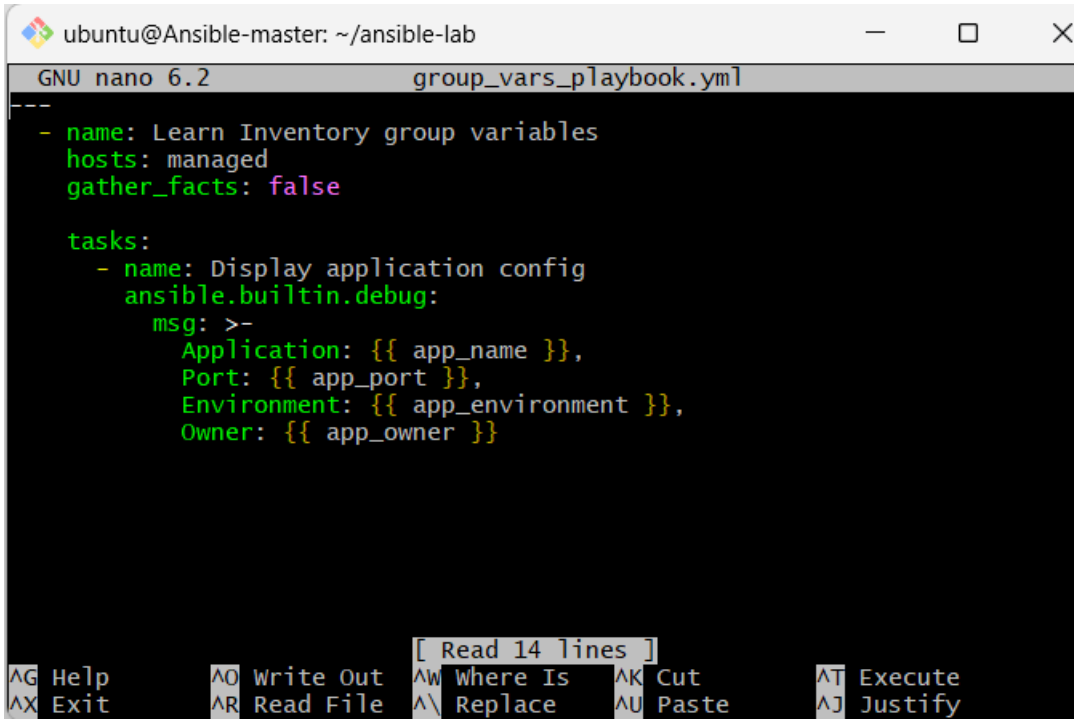
- name: Load variables from an external file
hosts: managed
gather_facts: false
vars_files: app_vars.yml
tasks:
- name: Display app config
ansible.builtin.debug:
msg: "Application {{ app_name }} uses {{ app_port }} in env {{ app_environment }}"
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line

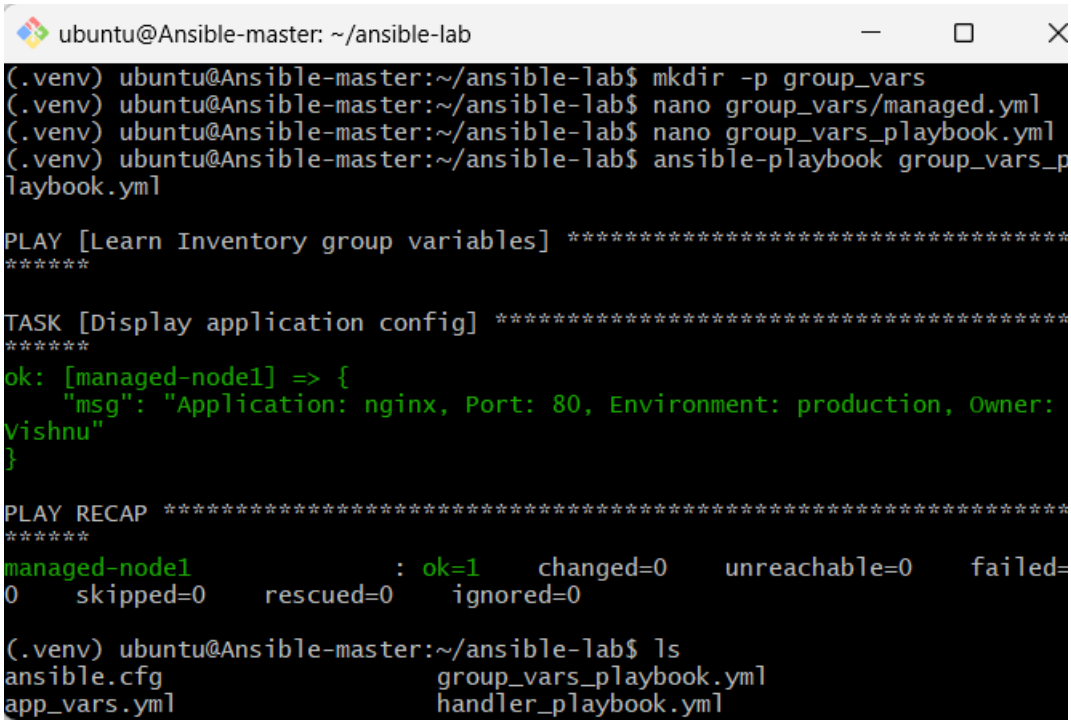

ubuntu@Ansible-master: ~/ansible-lab
(.venv) ubuntu@Ansible-master:~/ansible-lab$ ansible-playbook --syntax-che
iables_file_playbook.yml
playbook: variables_file_playbook.yml
(.venv) ubuntu@Ansible-master:~/ansible-lab$ eval "$(ssh-agent -s)"
Agent pid 873
(.venv) ubuntu@Ansible-master:~/ansible-lab$ ssh-add ansible_lab_key
ansible_lab_key: No such file or directory
(.venv) ubuntu@Ansible-master:~/ansible-lab$ ssh-add ~/.ssh/ansible_lab_ke
Enter passphrase for /home/ubuntu/.ssh/ansible_lab_key:
Identity added: /home/ubuntu/.ssh/ansible_lab_key (ubuntu@Ansible-master)
(.venv) ubuntu@Ansible-master:~/ansible-lab$ ansible-playbook variables_fi
ybook.yml
PLAY [Load variables from an external file] ****
TASK [Display app config] ****
ok: [managed-node1] => {
"msg": "Application nginx uses 80 in env development"
}
PLAY RECAP ****
managed-node1 : ok=1 changed=0 unreachable=0 failed=
kipped=0 rescued=0 ignored=0

Organizing variables using group_vars:

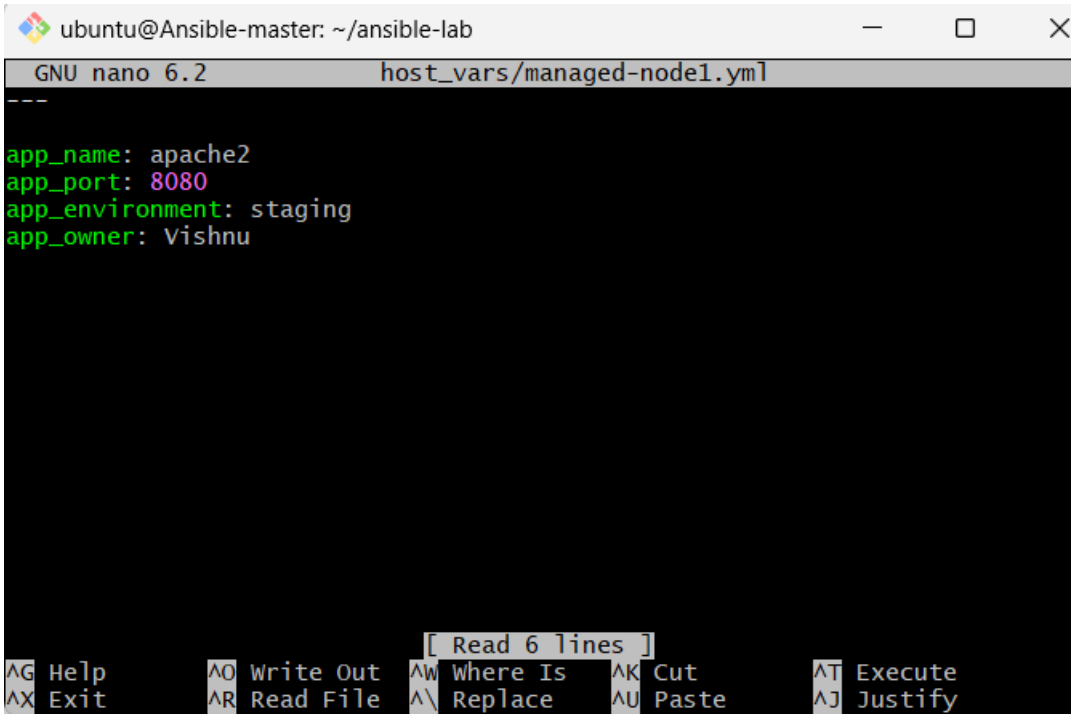


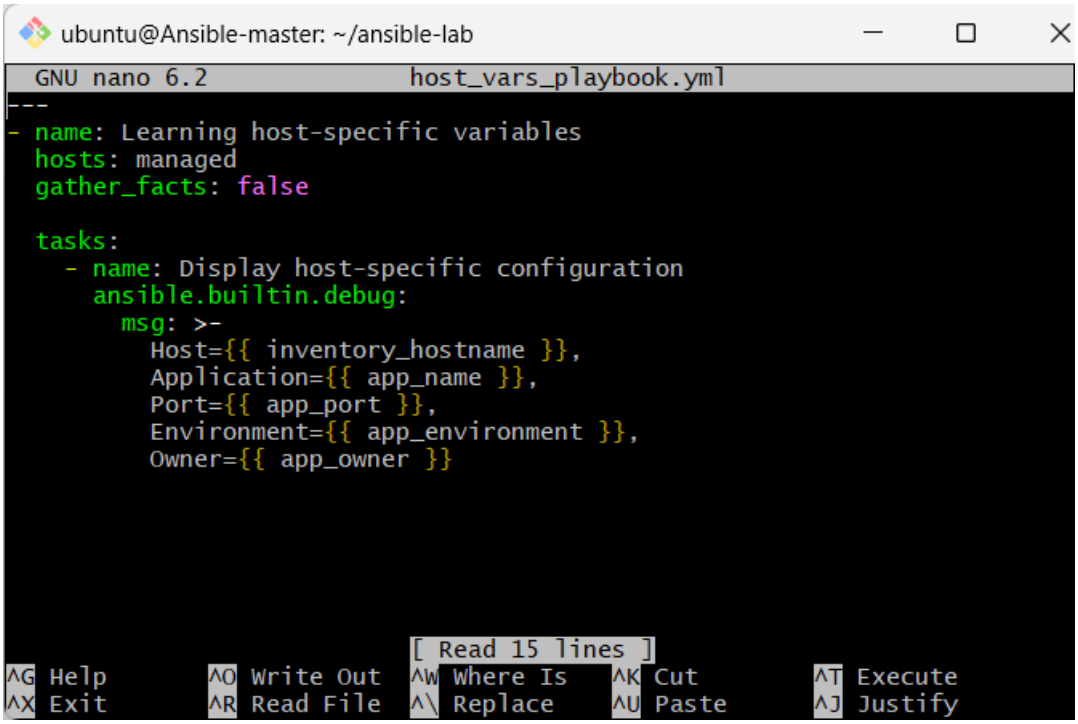
Note: Name the file after the inventory group: group_vars/managed.yml → group managed.

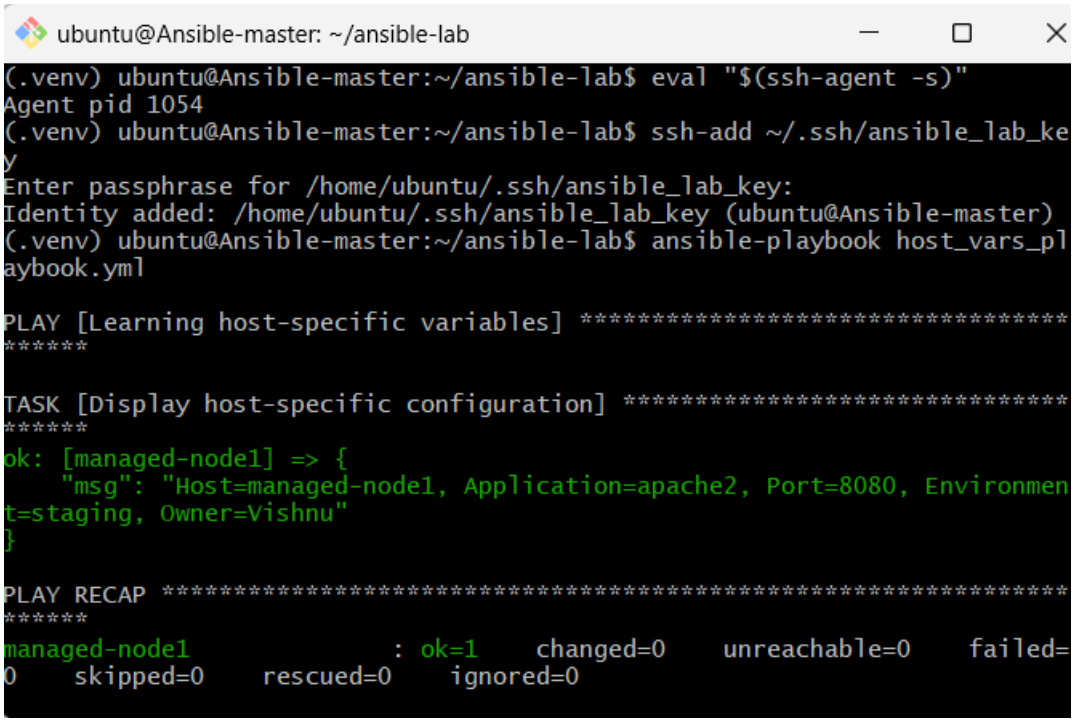




Organizing host-specific variables using host_vars:

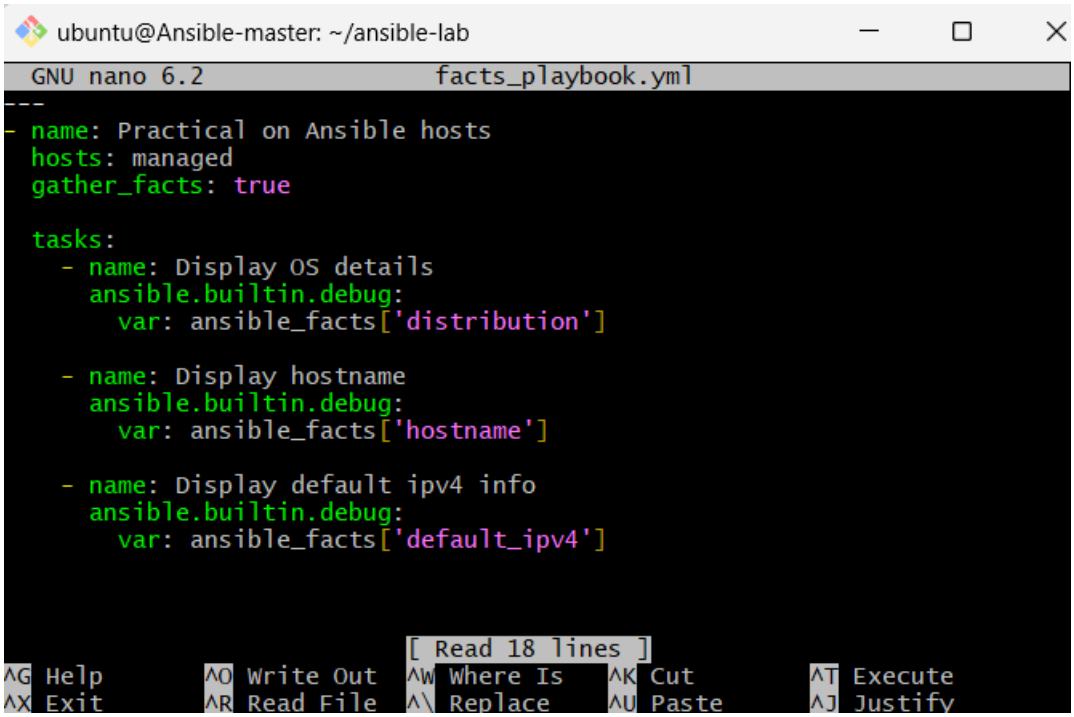


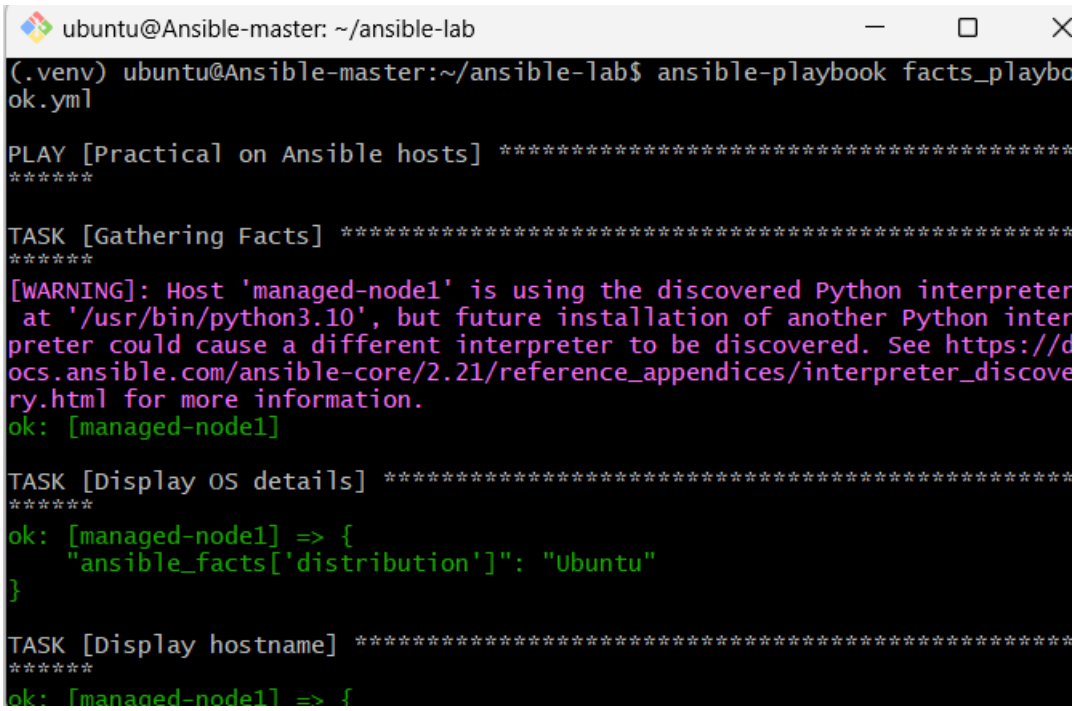


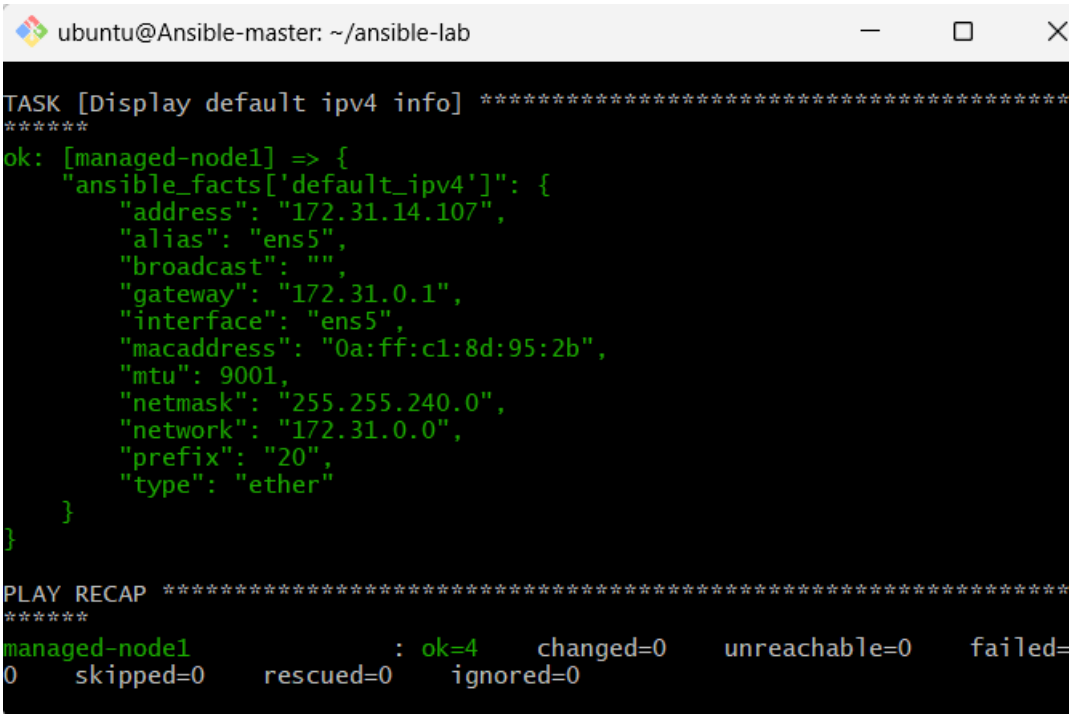


Note: Host variables (host_vars) override group variables (group_vars) when the same variable is defined in both locations.

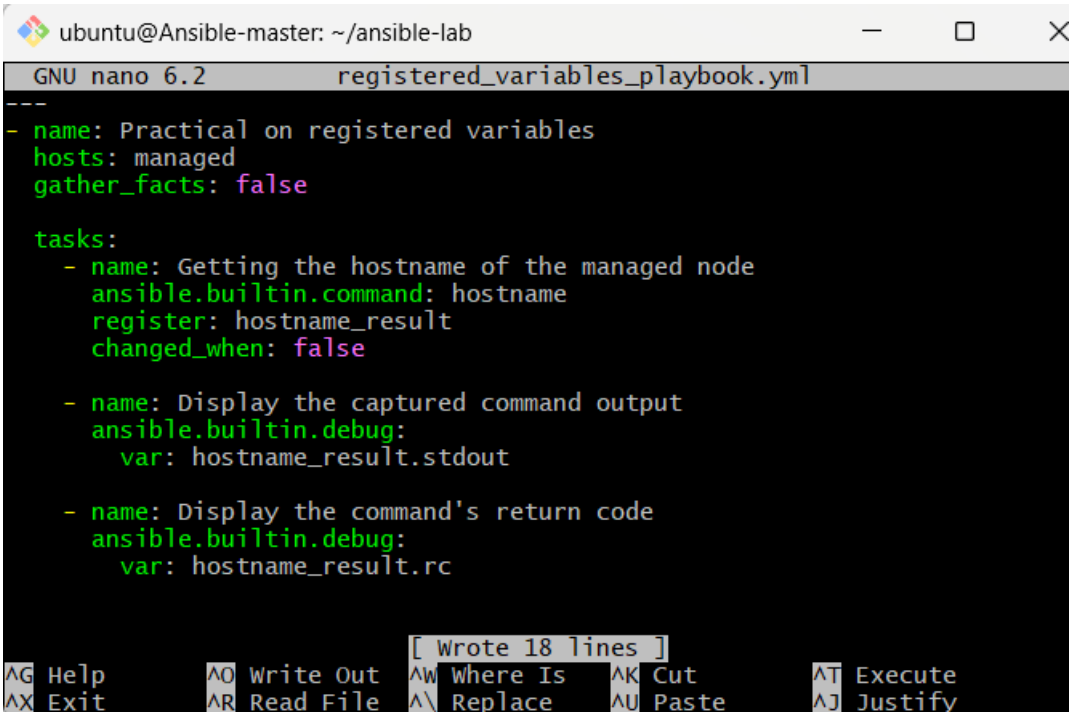
Ansible Facts:

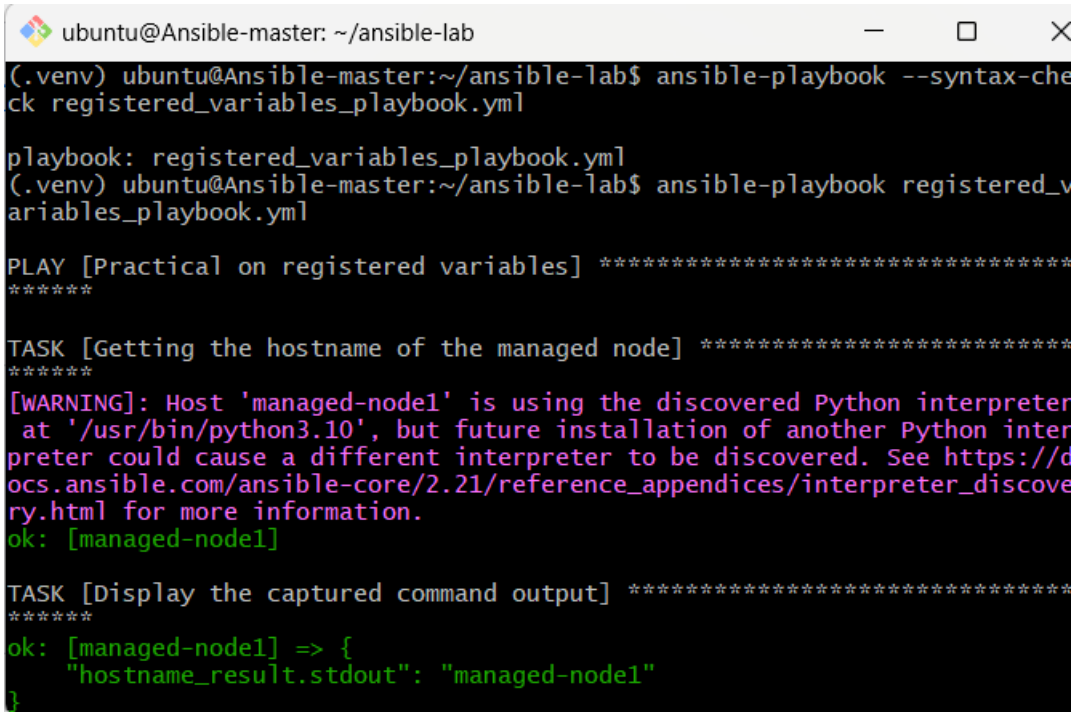


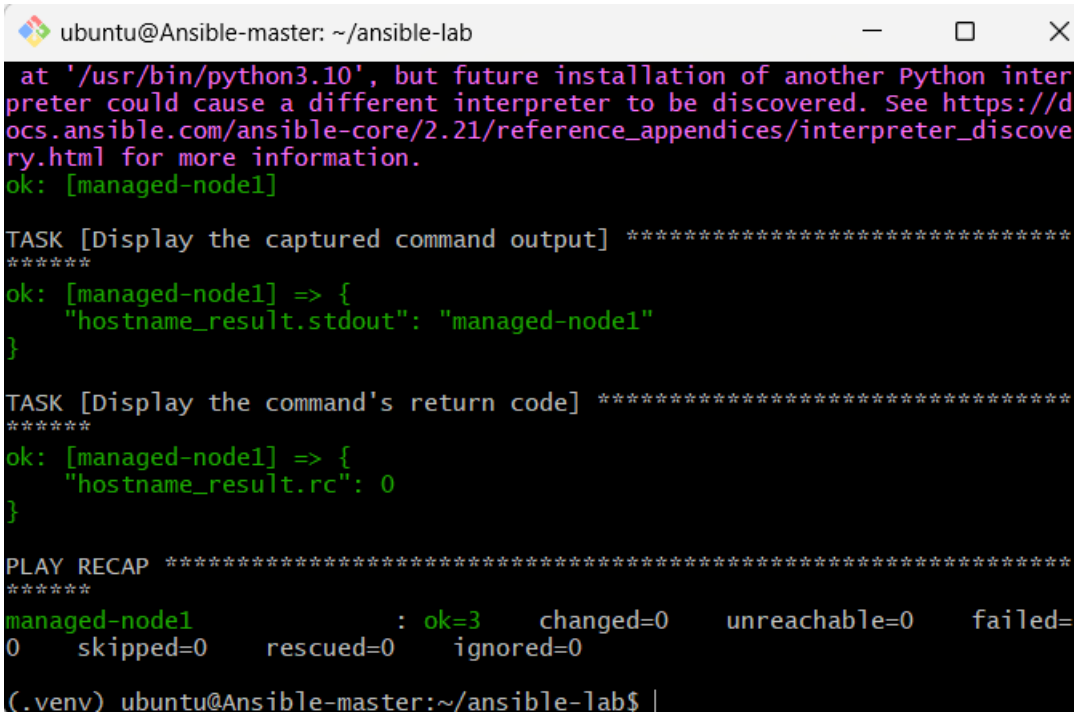




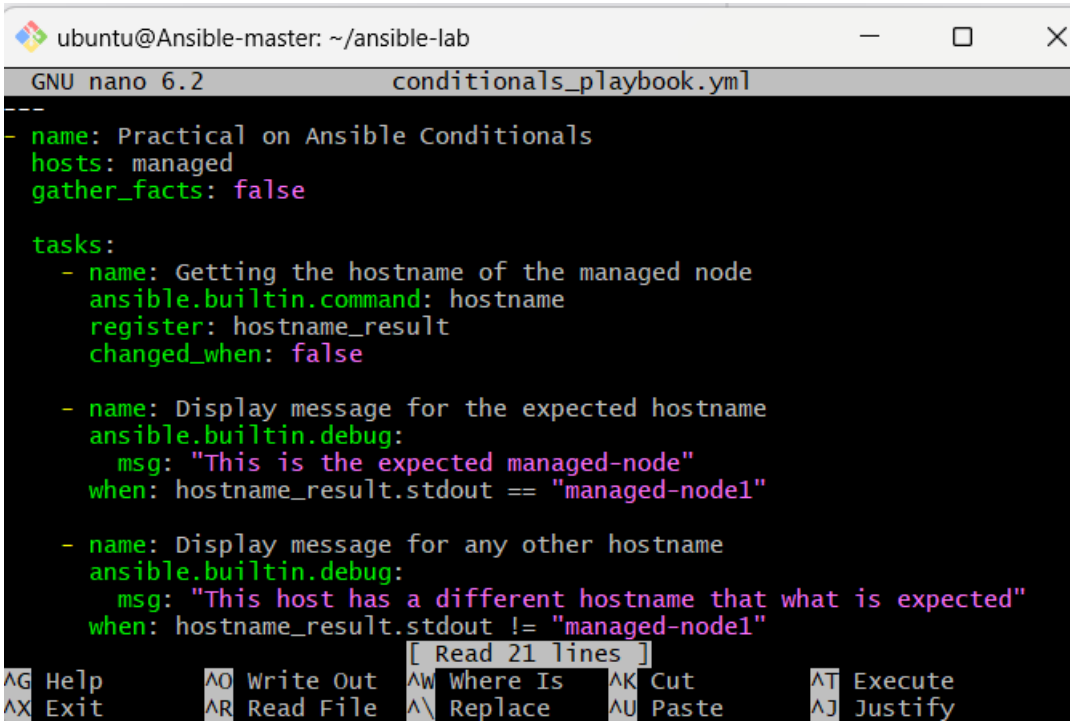
Registered variables:

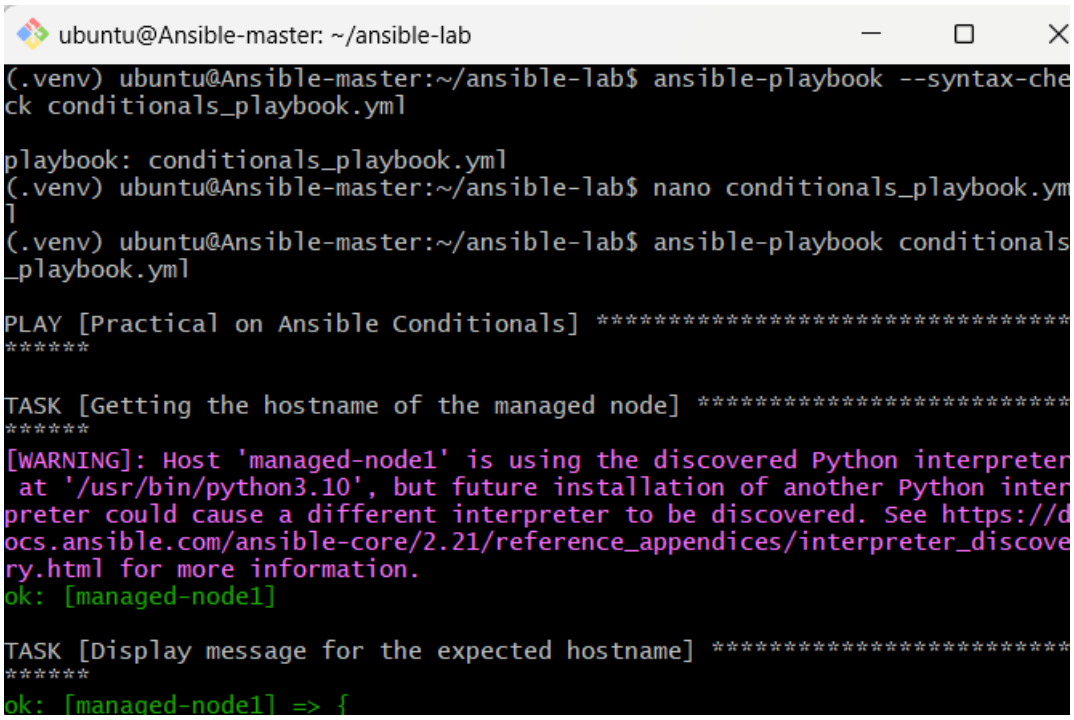


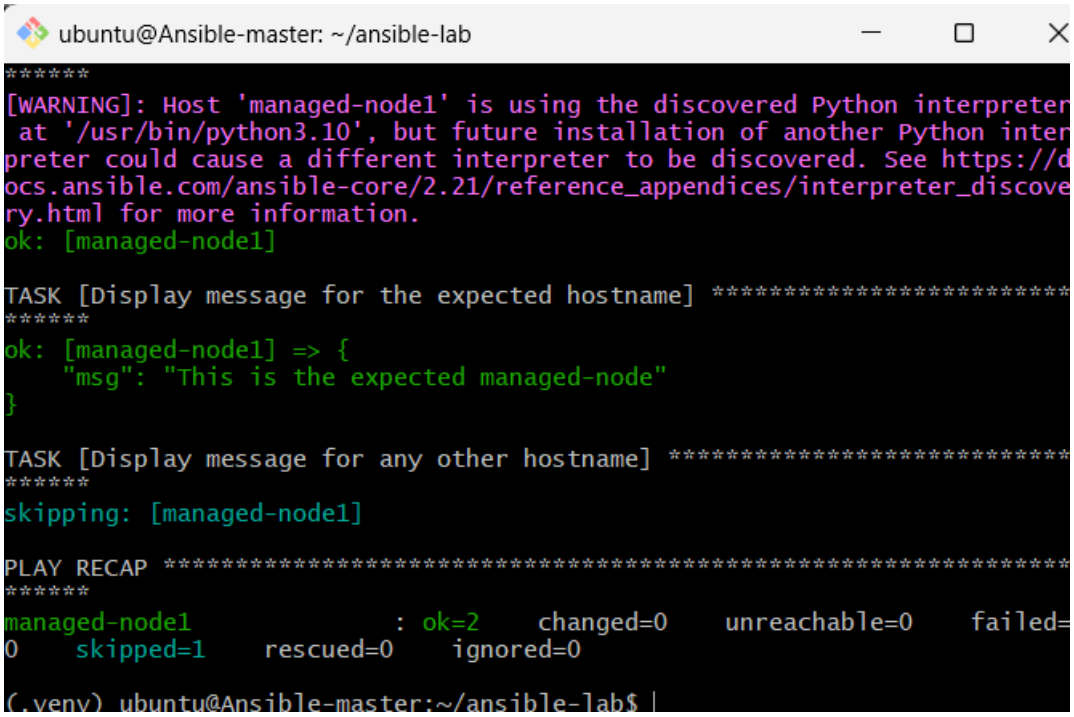

ubuntu@Ansible-master: ~/ansible-lab
(.venv) ubuntu@Ansible-master:~/ansible-lab$ ansible-playbook --syntax-che
ck registered_variables_playbook.yml
playbook: registered_variables_playbook.yml
(.venv) ubuntu@Ansible-master:~/ansible-lab$ ansible-playbook registered_v
ariables_playbook.yml
PLAY [Practical on registered variables] ***********************************
******
TASK [Getting the hostname of the managed node] ****************************
******
[WARNING]: Host 'managed-node1' is using the discovered Python interpreter
at '/usr/bin/python3.10', but future installation of another Python inter
preter could cause a different interpreter to be discovered. See https://d
ocs.ansible.com/ansible-core/2.21/reference_appendices/interpreter_discove
ry.html for more information.
ok: [managed-node1]
TASK [Display the captured command output] *********************************
******
ok: [managed-node1] => {
"hostname_result.stdout": "managed-node1"
}


ubuntu@Ansible-master: ~/ansible-lab
at '/usr/bin/python3.10', but future installation of another Python inter
preter could cause a different interpreter to be discovered. See https://d
ocs.ansible.com/ansible-core/2.21/reference_appendices/interpreter_discove
ry.html for more information.
ok: [managed-node1]
TASK [Display the captured command output] *********************************
******
ok: [managed-node1] => {
"hostname_result.stdout": "managed-node1"
}
TASK [Display the command's return code] ***********************************
******
ok: [managed-node1] => {
"hostname_result.rc": 0
}
PLAY RECAP ***
******
managed-node1 : ok=3 changed=0 unreachable=0 failed=
0 skipped=0 rescued=0 ignored=0
(.venv) ubuntu@Ansible-master:~/ansible-lab$

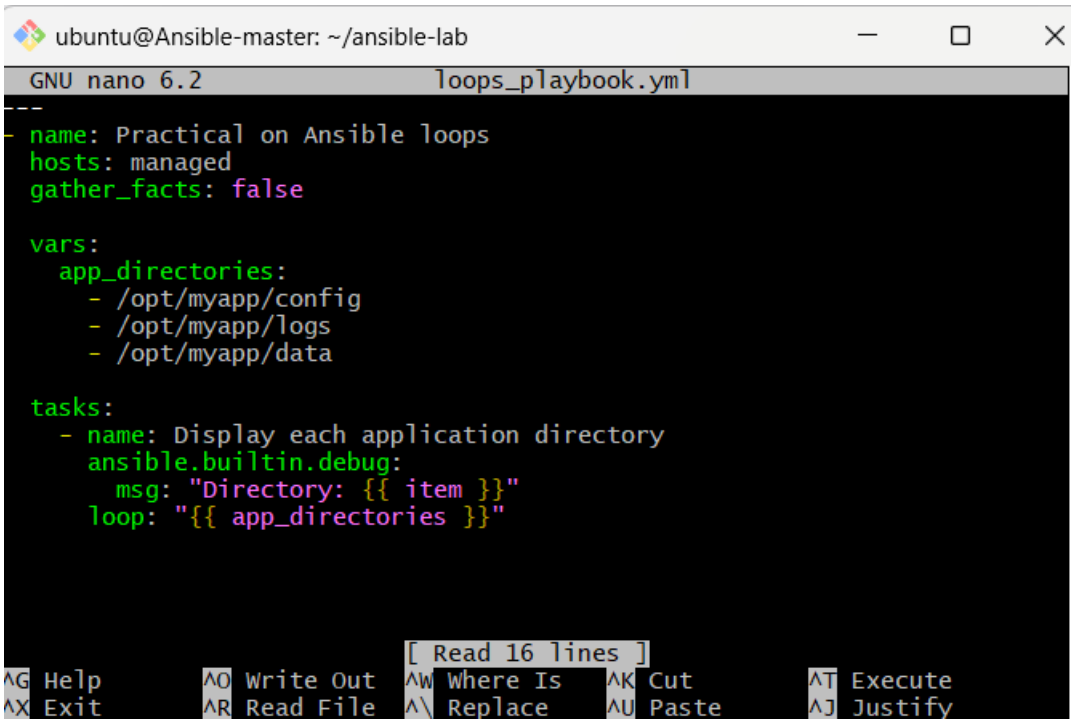
Conditionals using when:

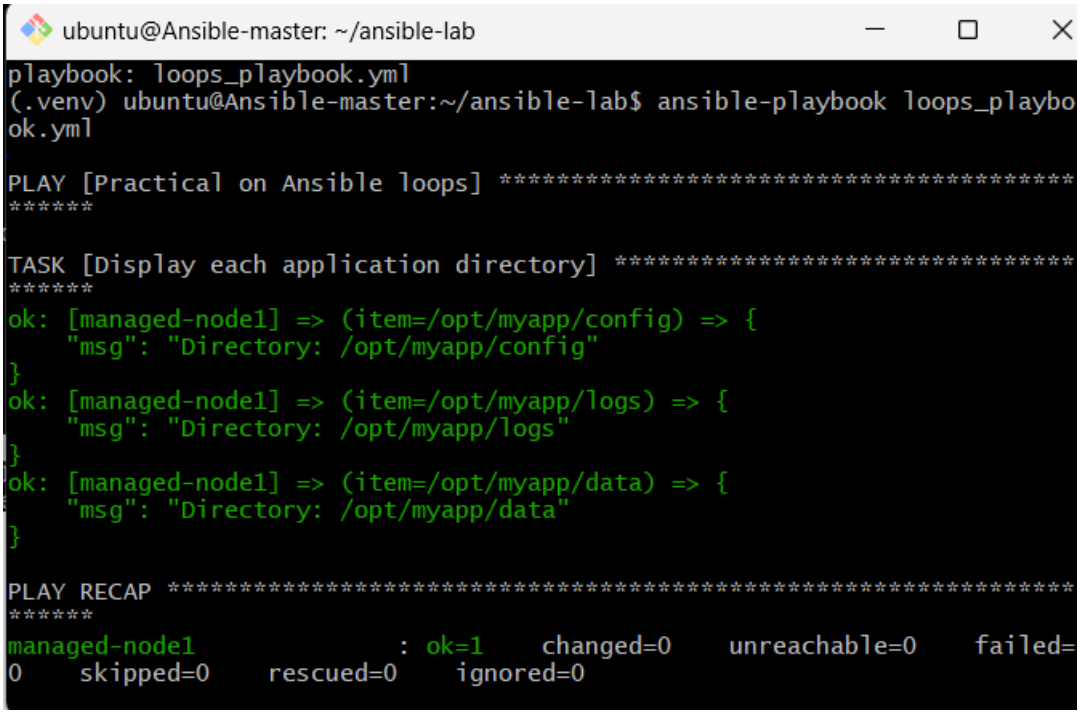


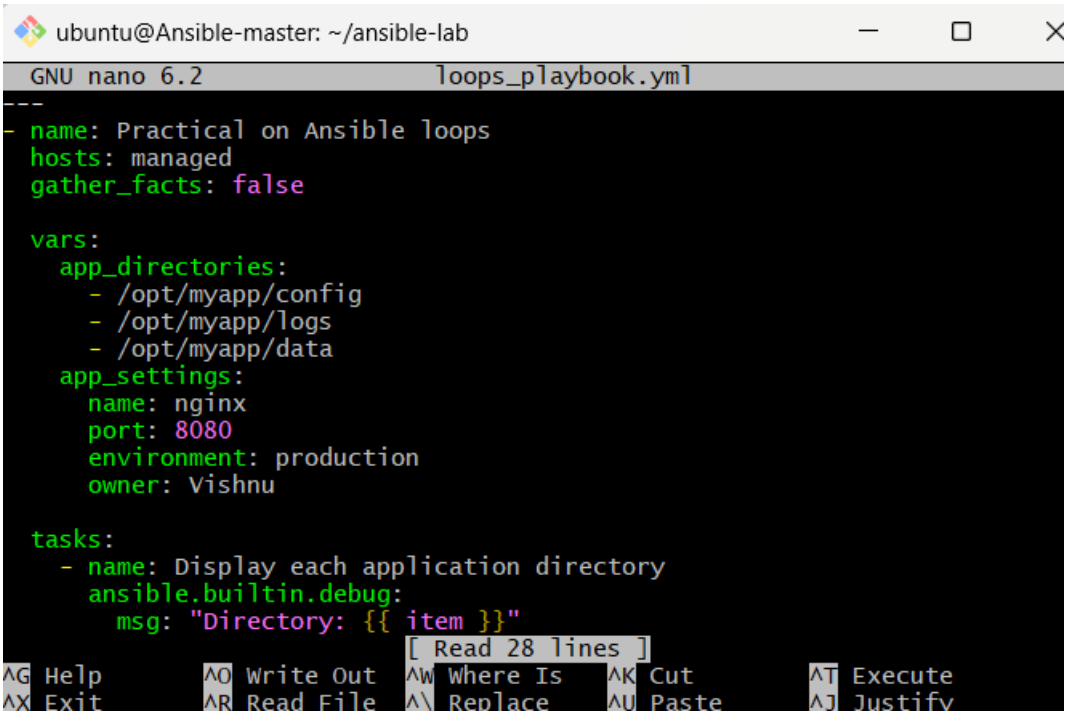




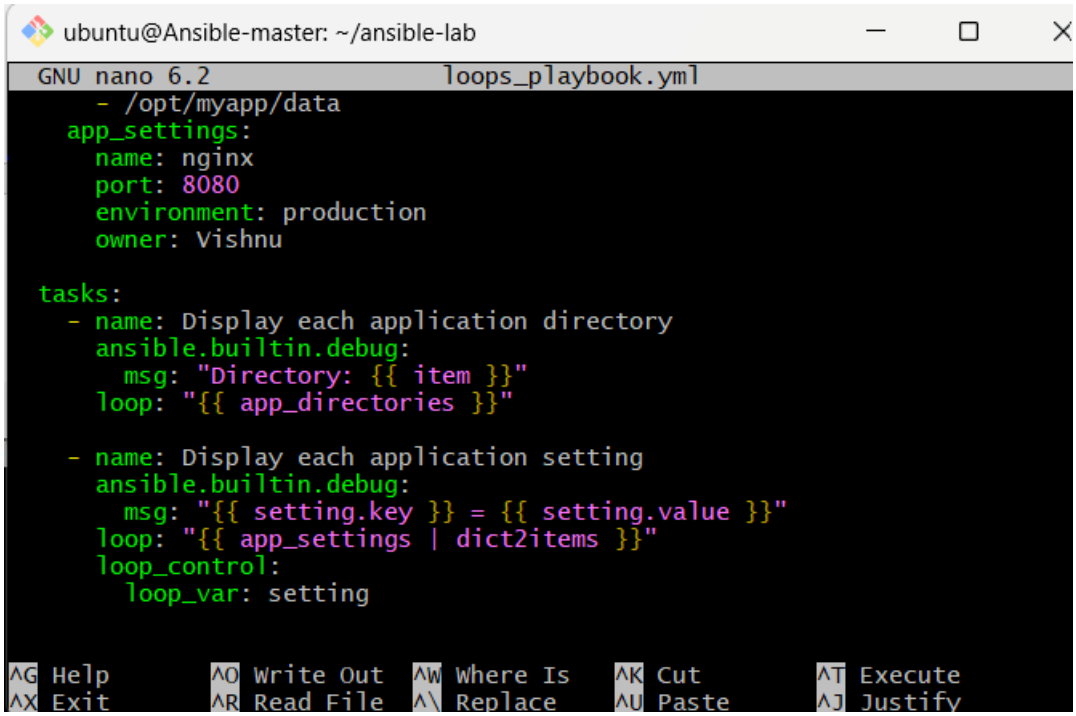
Ansible Loops:

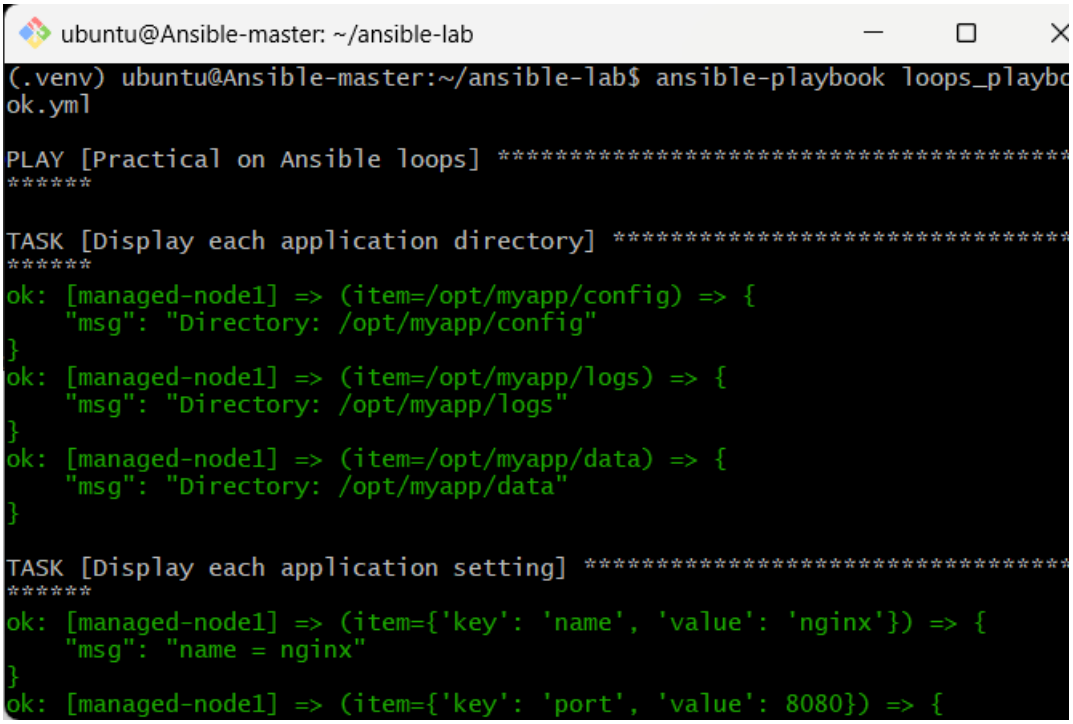


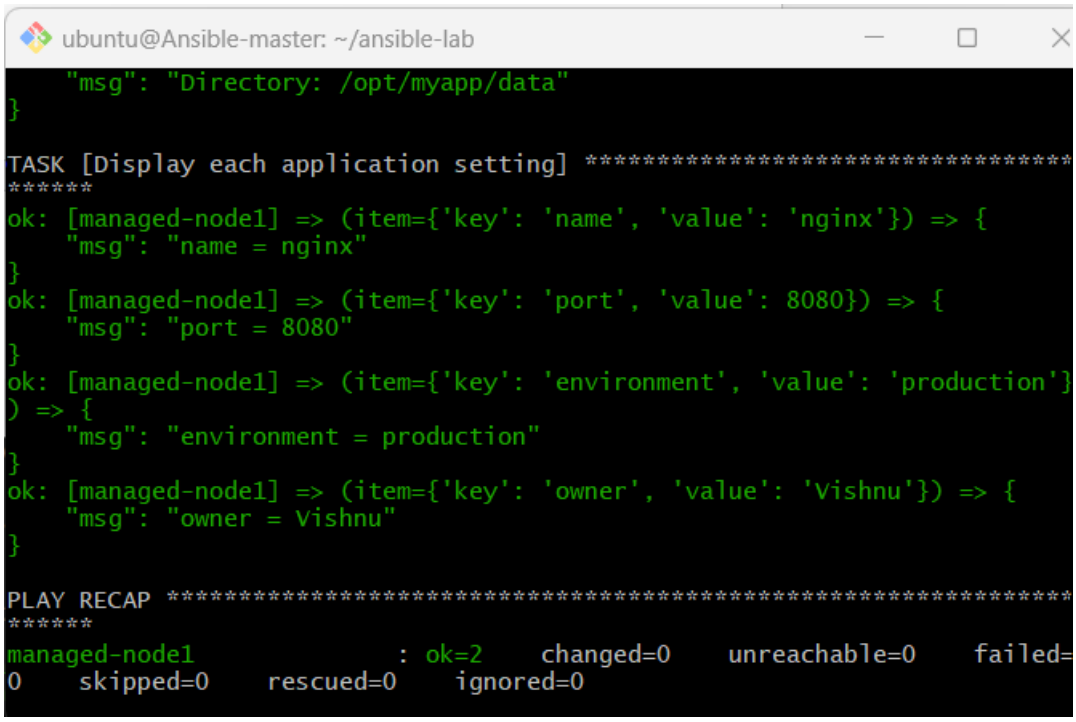

ubuntu@Ansible-master: ~/ansible-lab
playbook: loops_playbook.yml
(.venv) ubuntu@Ansible-master:~/ansible-lab$ ansible-playbook loops_playbo
ok.yml
PLAY [Practical on Ansible loops] **********
******
TASK [Display each application directory] **********
******
ok: [managed-node1] => (item=/opt/myapp/config) => {
"msg": "Directory: /opt/myapp/config"
}
ok: [managed-node1] => (item=/opt/myapp/logs) => {
"msg": "Directory: /opt/myapp/logs"
}
ok: [managed-node1] => (item=/opt/myapp/data) => {
"msg": "Directory: /opt/myapp/data"
}
PLAY RECAP **********
******
managed-node1 : ok=1 changed=0 unreachable=0 failed=
0 skipped=0 rescued=0 ignored=0


ubuntu@Ansible-master: ~/ansible-lab
GNU nano 6.2 loops_playbook.yml

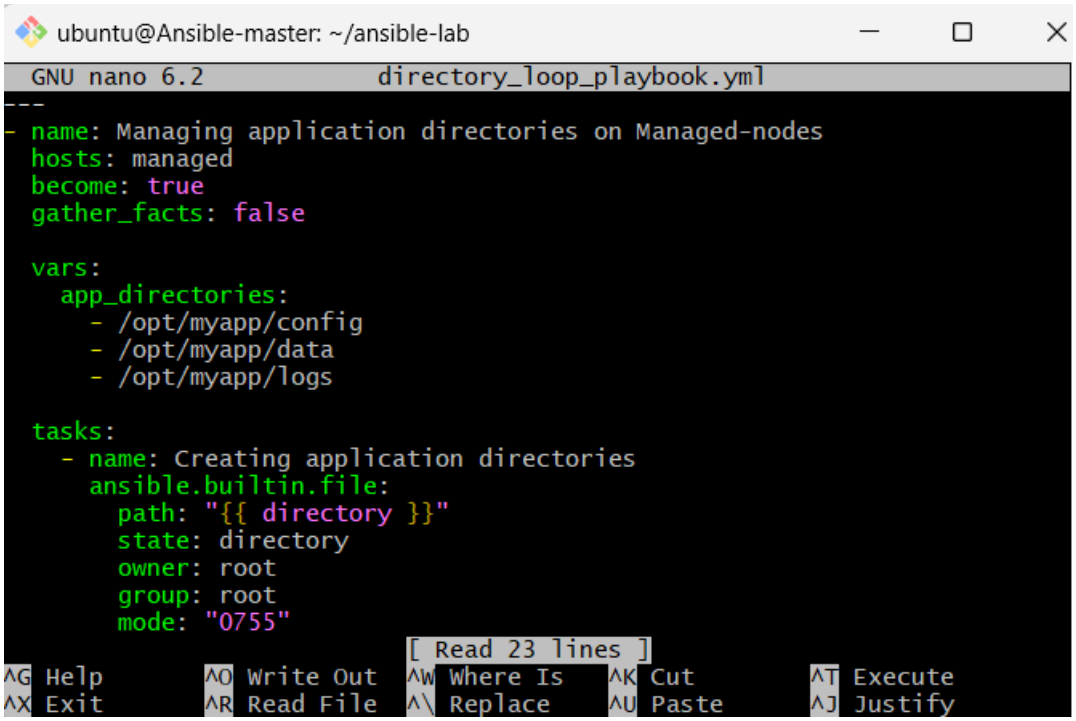
- name: Practical on Ansible loops
hosts: managed
gather_facts: false
vars:
app_directories:
- /opt/myapp/config
- /opt/myapp/logs
- /opt/myapp/data
app_settings:
name: nginx
port: 8080
environment: production
owner: Vishnu
tasks:
- name: Display each application directory
ansible.builtin.debug:
msg: "Directory: {{ item }}"
[Read 28 lines]
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

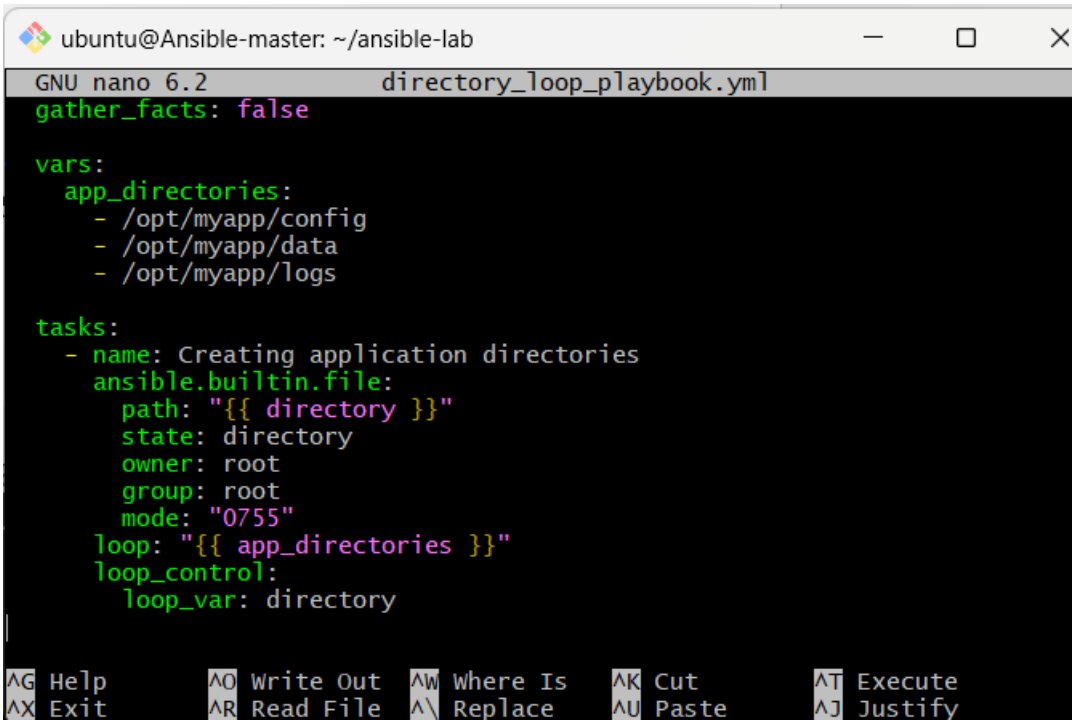

ubuntu@Ansible-master: ~/ansible-lab
GNU nano 6.2 loops_playbook.yml
- /opt/myapp/data
app_settings:
name: nginx
port: 8080
environment: production
owner: Vishnu
tasks:
- name: Display each application directory
ansible.builtin.debug:
msg: "Directory: {{ item }}"
loop: "{{ app_directories }}"
- name: Display each application setting
ansible.builtin.debug:
msg: "{{ setting.key }} = {{ setting.value }}"
loop: "{{ app_settings | dict2items }}"
loop_control:
loop_var: setting
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

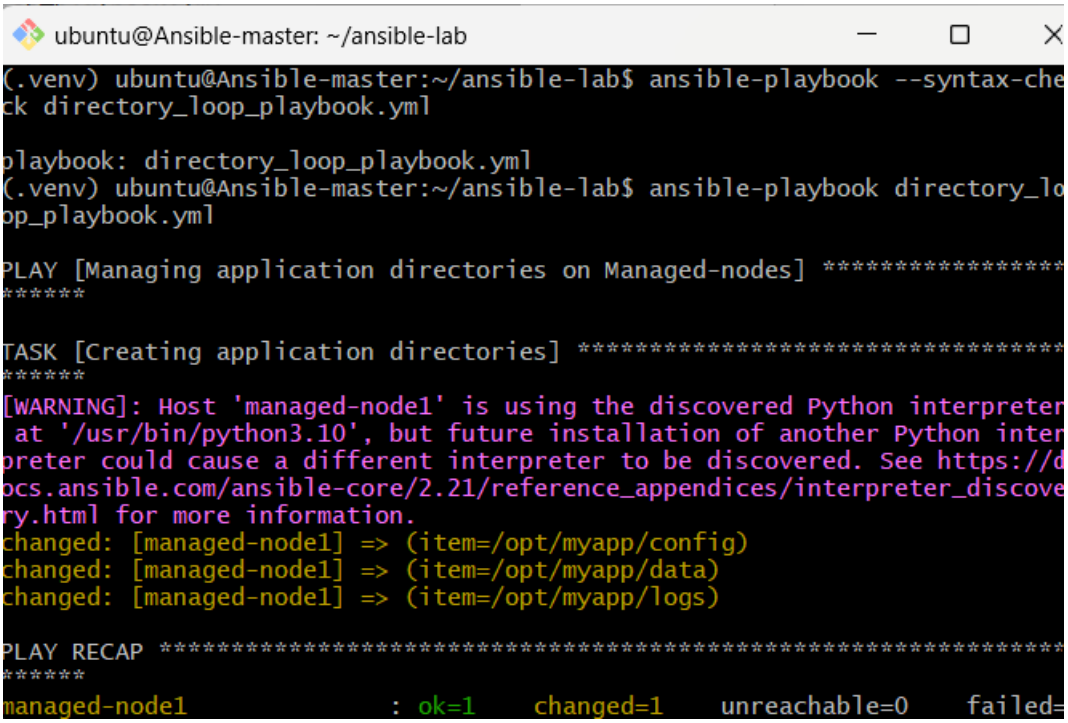

ubuntu@Ansible-master: ~/ansible-lab
(.venv) ubuntu@Ansible-master:~/ansible-lab$ ansible-playbook loops_playbo
ok.yml
PLAY [Practical on Ansible loops] **
******
TASK [Display each application directory] ********************************
******
ok: [managed-node1] => (item=/opt/myapp/config) => {
"msg": "Directory: /opt/myapp/config"
}
ok: [managed-node1] => (item=/opt/myapp/logs) => {
"msg": "Directory: /opt/myapp/logs"
}
ok: [managed-node1] => (item=/opt/myapp/data) => {
"msg": "Directory: /opt/myapp/data"
}
TASK [Display each application setting] ********************************
******
ok: [managed-node1] => (item={'key': 'name', 'value': 'nginx'}) => {
"msg": "name = nginx"
}
ok: [managed-node1] => (item={'key': 'port', 'value': 8080}) => {

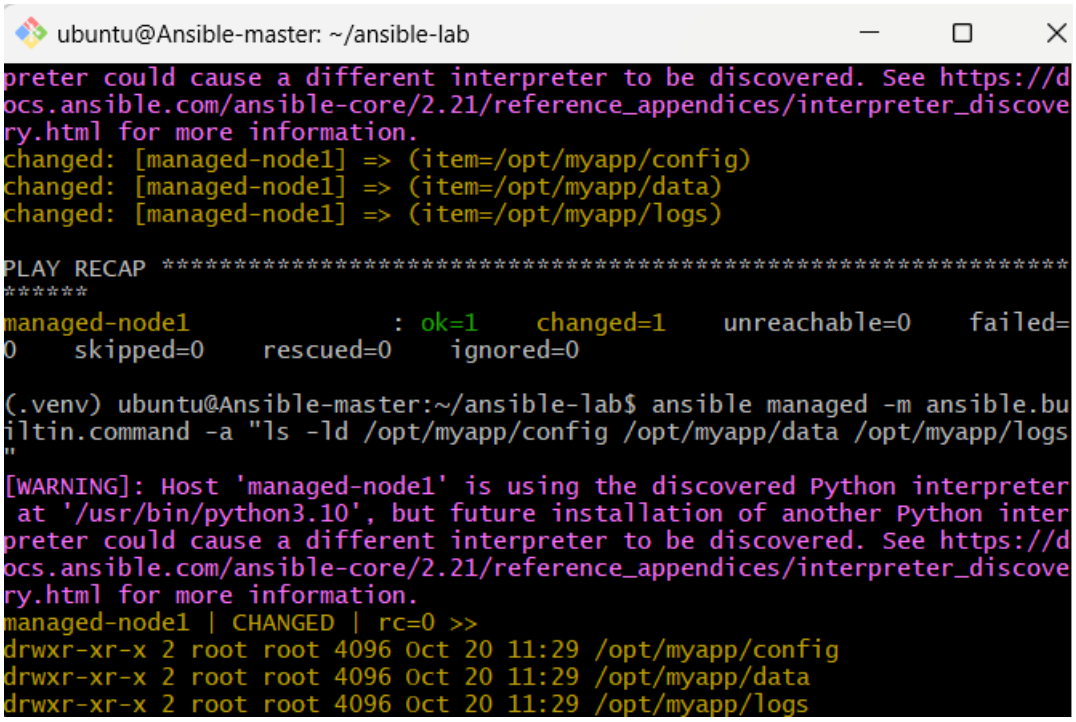


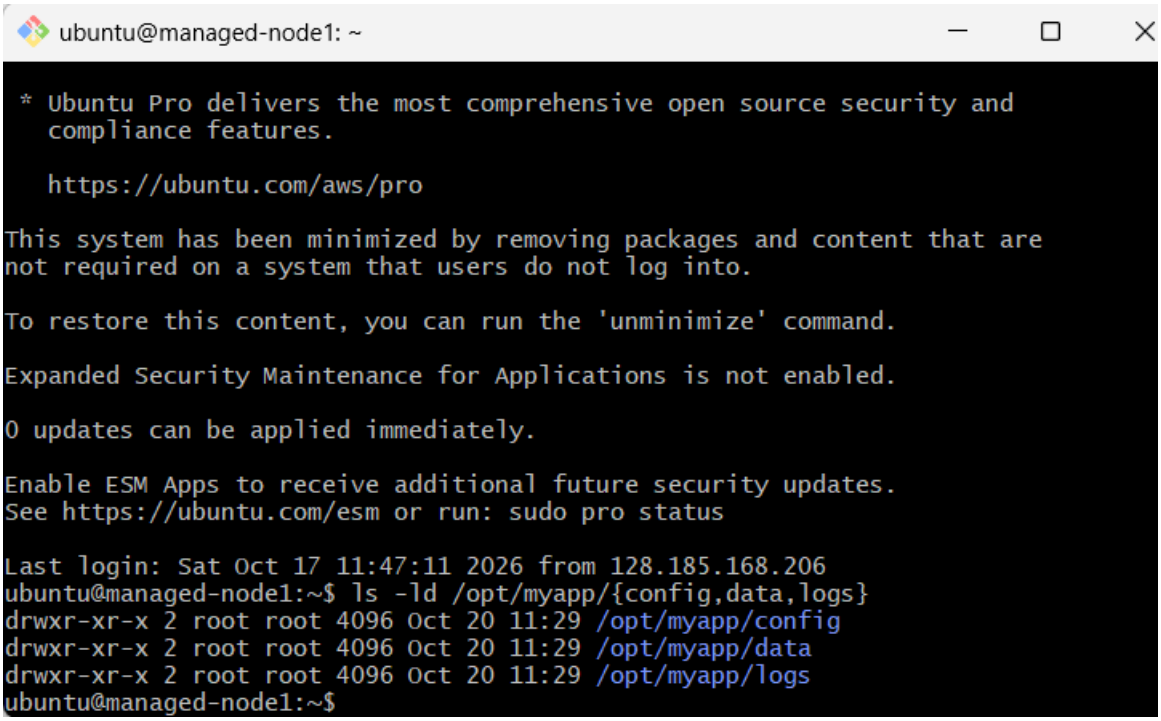
Creating directories on the managed-node using loops:



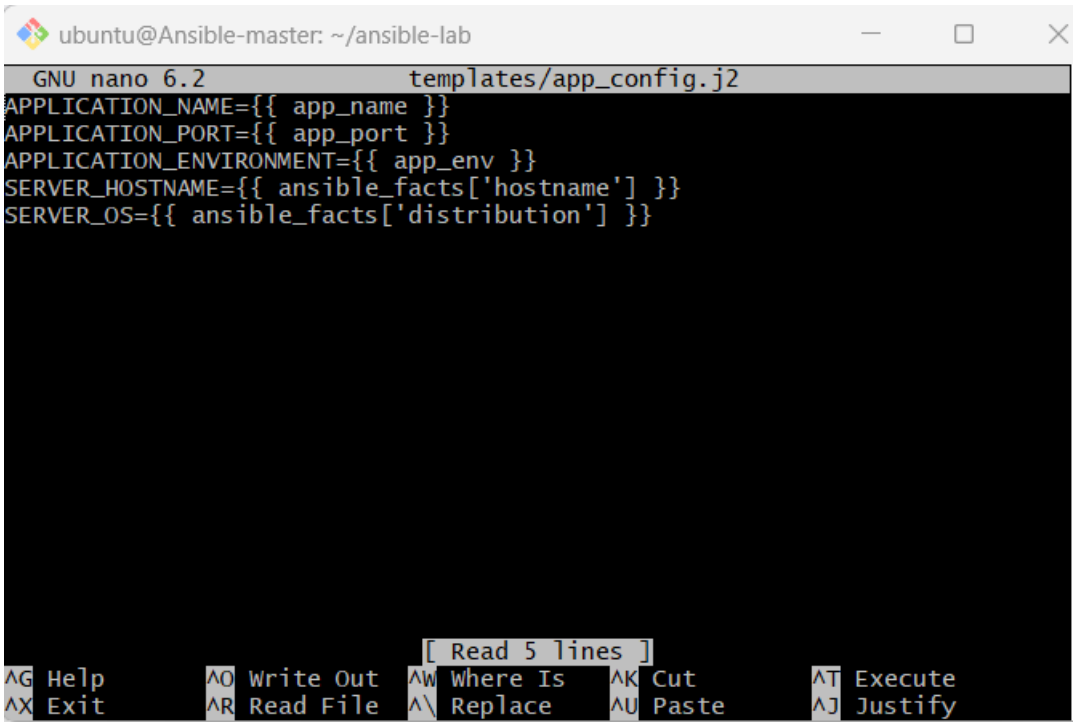

ubuntu@Ansible-master: ~/ansible-lab
GNU nano 6.2 directory_loop_playbook.yml
gather_facts: false
vars:
app_directories:
- /opt/myapp/config
- /opt/myapp/data
- /opt/myapp/logs
tasks:
- name: Creating application directories
ansible.builtin.file:
path: "{{ directory }}"
state: directory
owner: root
group: root
mode: "0755"
loop: "{{ app_directories }}"
loop_control:
loop_var: directory
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify

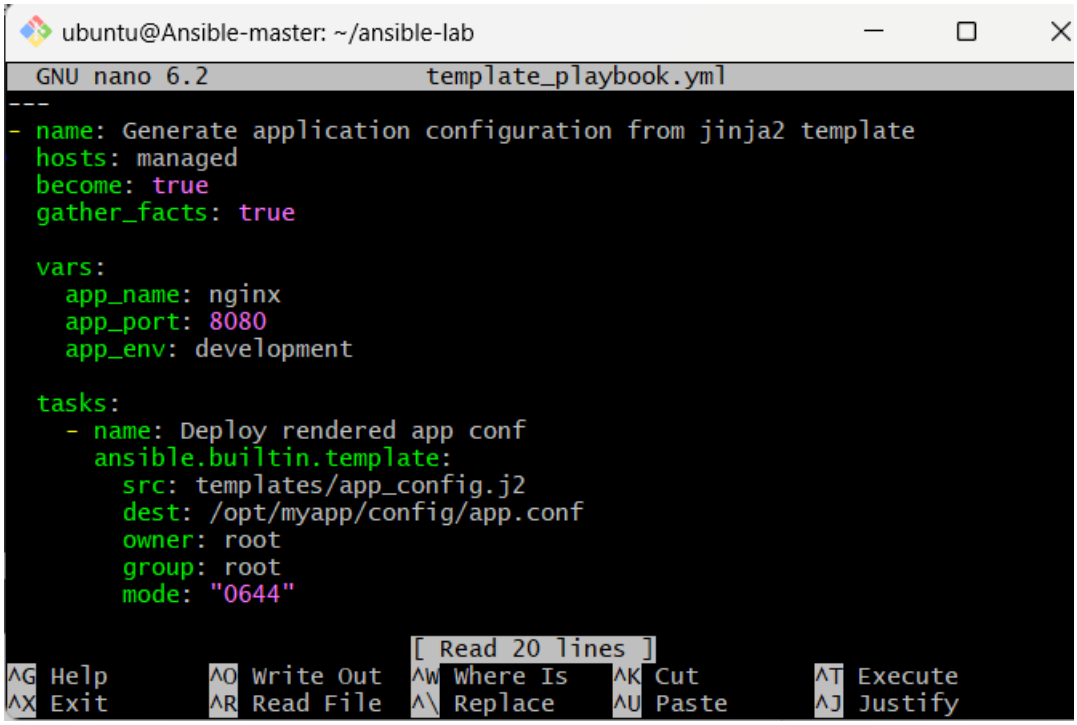

ubuntu@Ansible-master: ~/ansible-lab
(.venv) ubuntu@Ansible-master:~/ansible-lab$ ansible-playbook --syntax-che
ck directory_loop_playbook.yml
playbook: directory_loop_playbook.yml
(.venv) ubuntu@Ansible-master:~/ansible-lab$ ansible-playbook directory_lo
op_playbook.yml
PLAY [Managing application directories on Managed-nodes] ****************
******
TASK [Creating application directories] ********************************
******
[WARNING]: Host 'managed-node1' is using the discovered Python interpreter
at '/usr/bin/python3.10', but future installation of another Python inter
preter could cause a different interpreter to be discovered. See https://d
ocs.ansible.com/ansible-core/2.21/reference_appendices/interpreter_discove
ry.html for more information.
changed: [managed-node1] => (item=/opt/myapp/config)
changed: [managed-node1] => (item=/opt/myapp/data)
changed: [managed-node1] => (item=/opt/myapp/logs)
PLAY RECAP ***
******
managed-node1 : ok=1 changed=1 unreachable=0 failed=

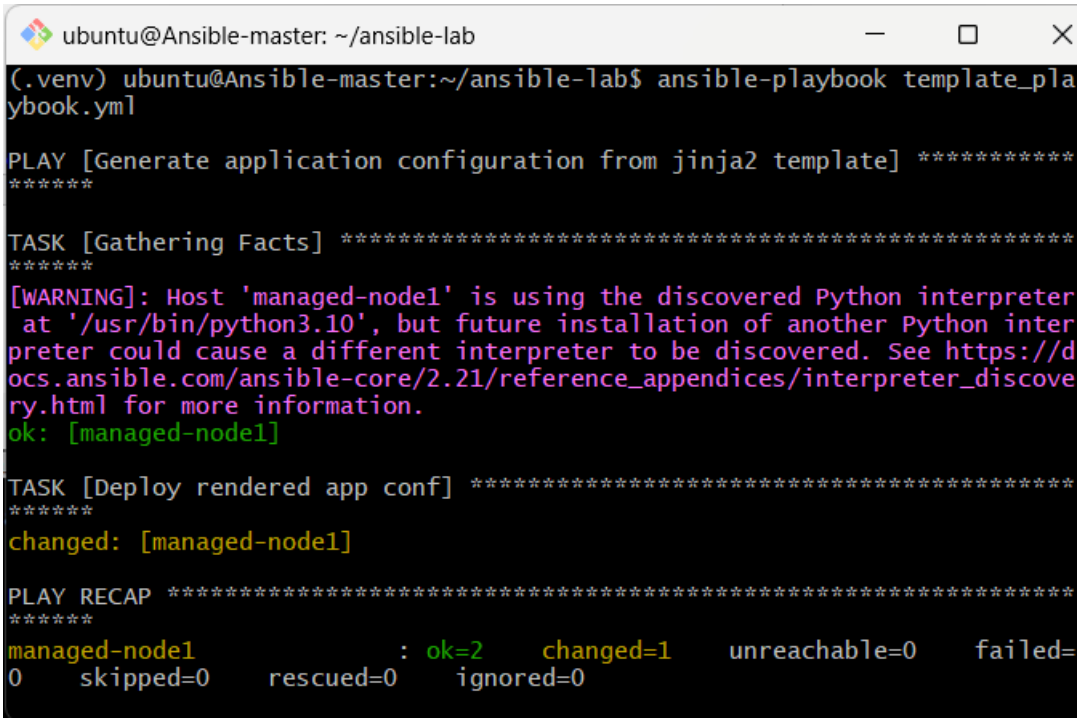

ubuntu@Ansible-master: ~/ansible-lab
preter could cause a different interpreter to be discovered. See https://d
ocs.ansible.com/ansible-core/2.21/reference_appendices/interpreter_discove
ry.html for more information.
changed: [managed-node1] => (item=/opt/myapp/config)
changed: [managed-node1] => (item=/opt/myapp/data)
changed: [managed-node1] => (item=/opt/myapp/logs)
PLAY RECAP
******
managed-node1 : ok=1 changed=1 unreachable=0 failed=
0 skipped=0 rescued=0 ignored=0
(.venv) ubuntu@Ansible-master:~/ansible-lab$ ansible managed -m ansible.bu
iltin.command -a "ls -ld /opt/myapp/config /opt/myapp/data /opt/myapp/logs
"
[WARNING]: Host 'managed-node1' is using the discovered Python interpreter
at '/usr/bin/python3.10', but future installation of another Python inter
preter could cause a different interpreter to be discovered. See https://d
ocs.ansible.com/ansible-core/2.21/reference_appendices/interpreter_discove
ry.html for more information.
managed-node1 | CHANGED | rc=0 >>
drwxr-xr-x 2 root root 4096 Oct 20 11:29 /opt/myapp/config
drwxr-xr-x 2 root root 4096 Oct 20 11:29 /opt/myapp/data
drwxr-xr-x 2 root root 4096 Oct 20 11:29 /opt/myapp/logs

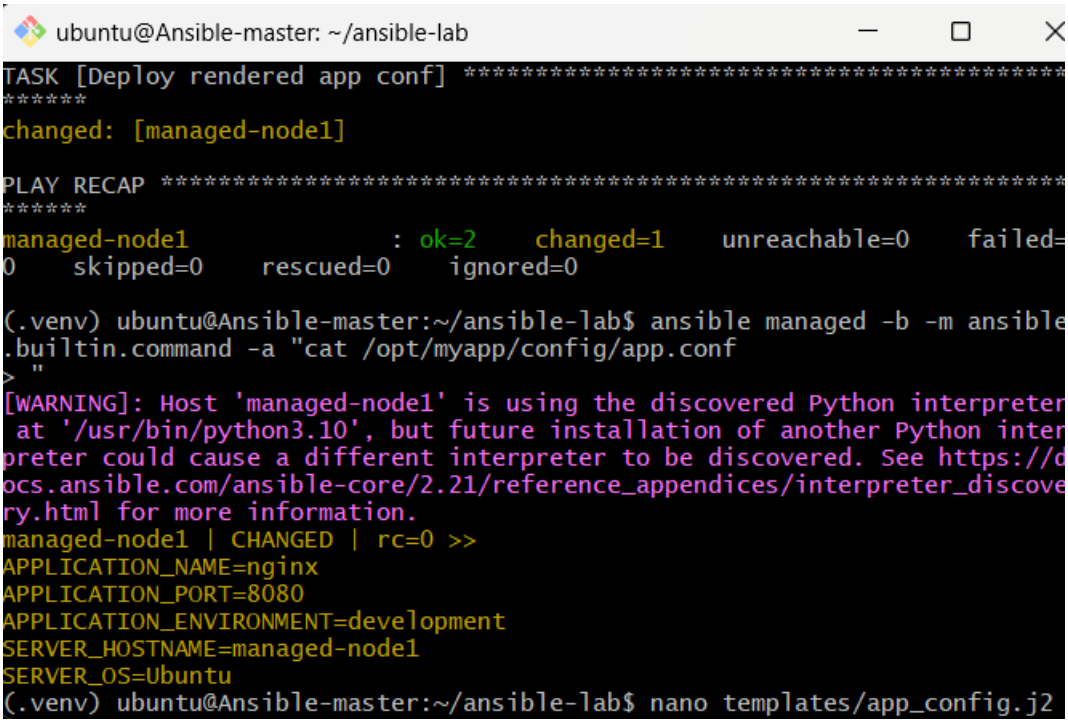

ubuntu@managed-node1: ~
* Ubuntu Pro delivers the most comprehensive open source security and
compliance features.
https://ubuntu.com/aws/pro
This system has been minimized by removing packages and content that are
not required on a system that users do not log into.
To restore this content, you can run the 'unminimize' command.
Expanded Security Maintenance for Applications is not enabled.
0 updates can be applied immediately.
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
Last login: Sat Oct 17 11:47:11 2026 from 128.185.168.206
ubuntu@managed-node1:~$ ls -ld /opt/myapp/{config,data,logs}
drwxr-xr-x 2 root root 4096 Oct 20 11:29 /opt/myapp/config
drwxr-xr-x 2 root root 4096 Oct 20 11:29 /opt/myapp/data
drwxr-xr-x 2 root root 4096 Oct 20 11:29 /opt/myapp/logs
ubuntu@managed-node1:~$

Jinja2 templates with Ansible:

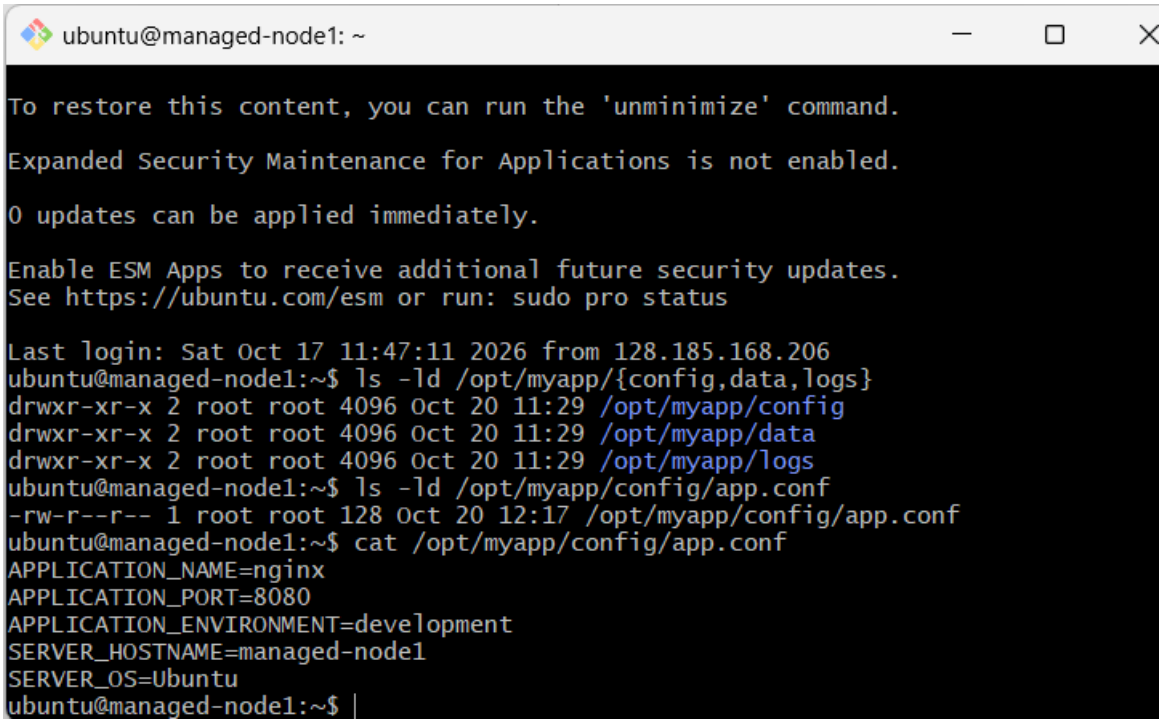




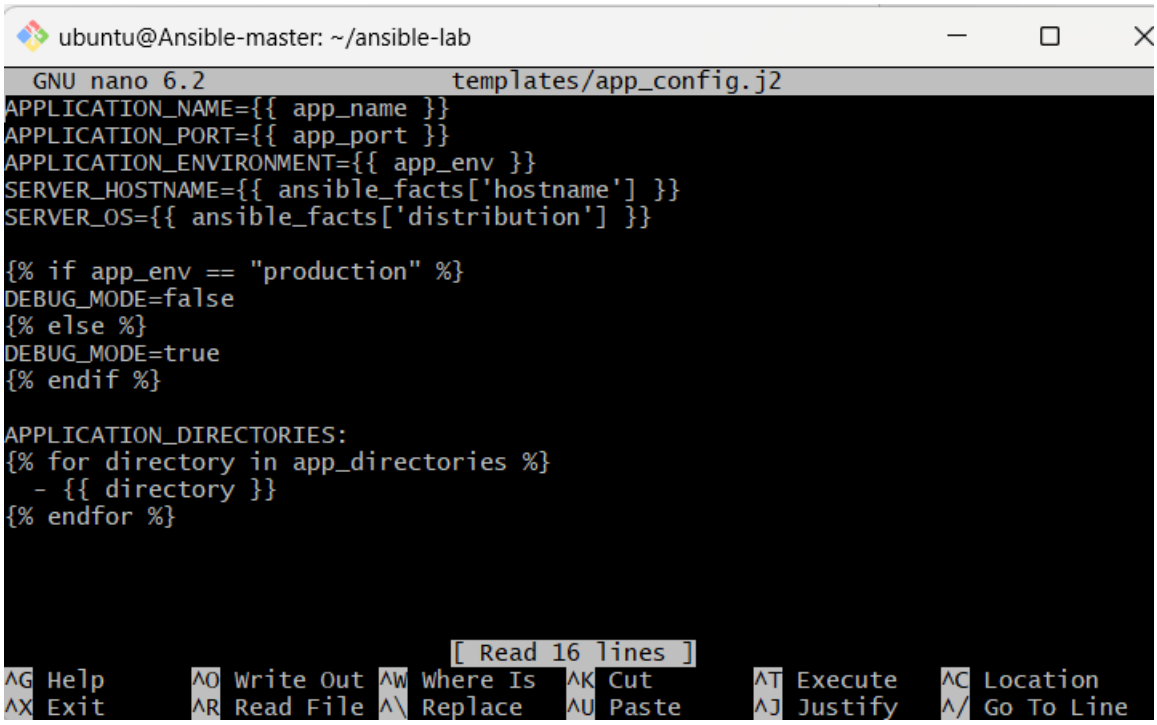


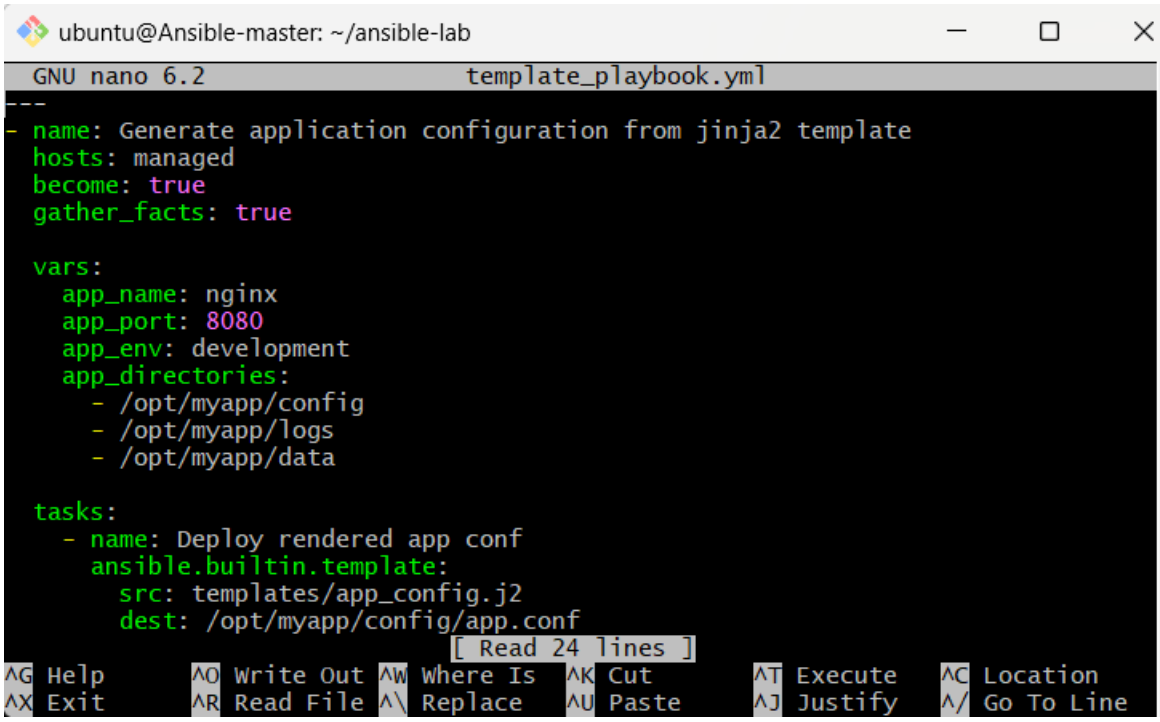


There we go, successfully rendered and deployed

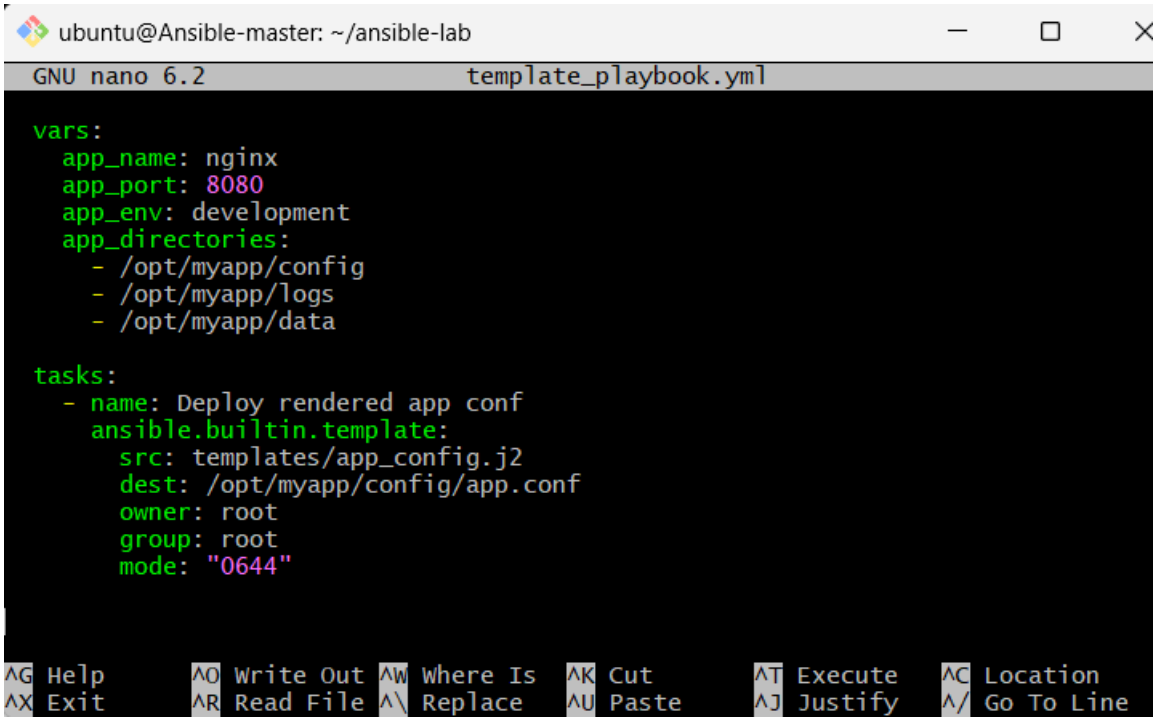


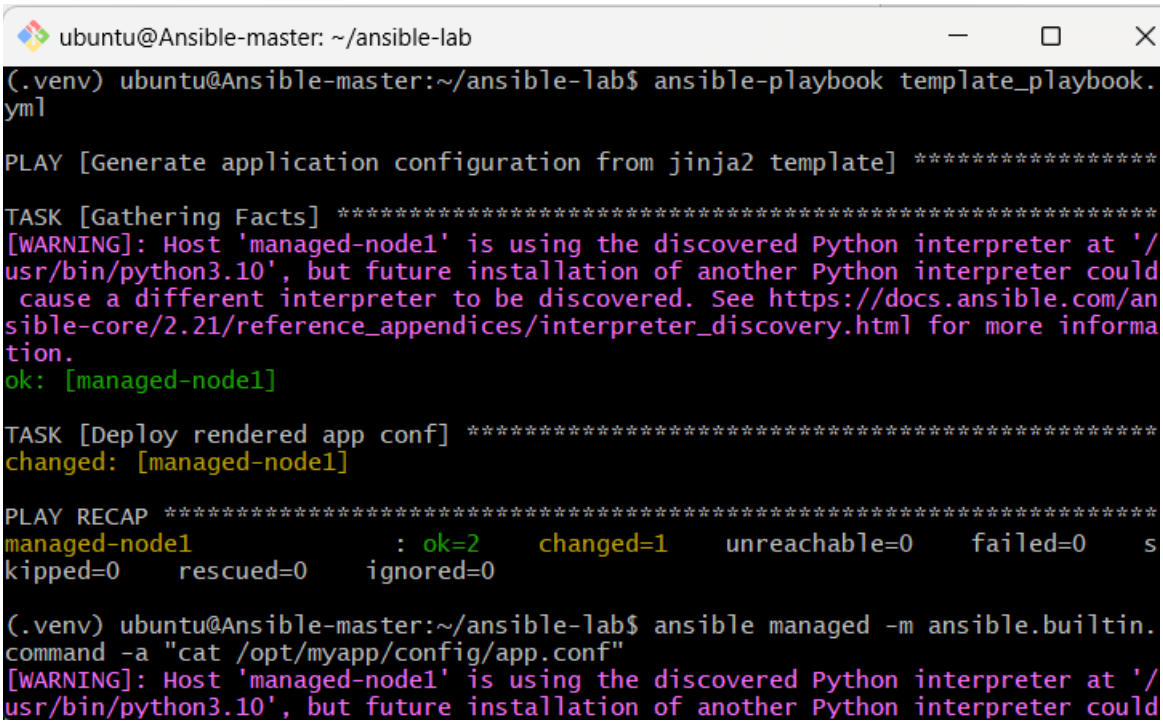
Jinja2 conditionals and loops inside templates

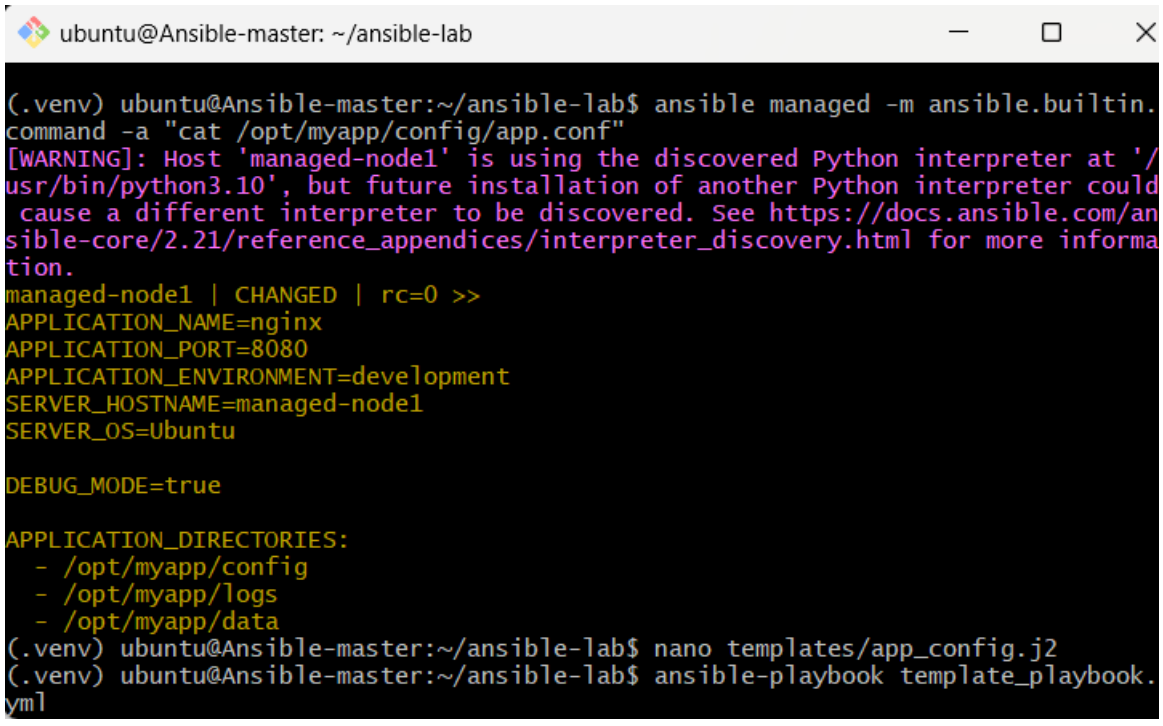


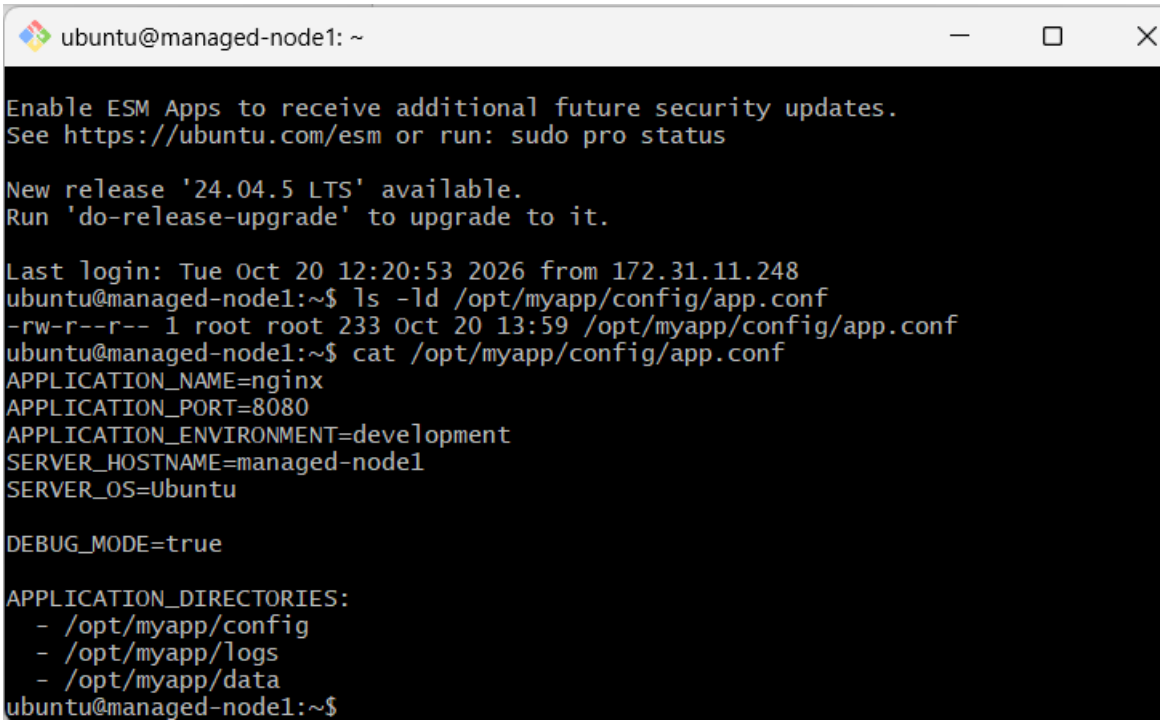

ubuntu@Ansible-master: ~/ansible-lab
GNU nano 6.2 template_playbook.yml

- name: Generate application configuration from jinja2 template
hosts: managed
become: true
gather_facts: true
vars:
app_name: nginx
app_port: 8080
app_env: development
app_directories:
- /opt/myapp/config
- /opt/myapp/logs
- /opt/myapp/data
tasks:
- name: Deploy rendered app conf
ansible.builtin.template:
src: templates/app_config.j2
dest: /opt/myapp/config/app.conf
[Read 24 lines]
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line


ubuntu@Ansible-master: ~/ansible-lab
GNU nano 6.2 template_playbook.yml
vars:
app_name: nginx
app_port: 8080
app_env: development
app_directories:
- /opt/myapp/config
- /opt/myapp/logs
- /opt/myapp/data
tasks:
- name: Deploy rendered app conf
ansible.builtin.template:
src: templates/app_config.j2
dest: /opt/myapp/config/app.conf
owner: root
group: root
mode: "0644"
^G Help ^O Write Out ^W Where Is ^K Cut ^T Execute ^C Location
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^/ Go To Line


ubuntu@Ansible-master: ~/ansible-lab
(.venv) ubuntu@Ansible-master:~/ansible-lab$ ansible-playbook template_playbook.yml
PLAY [Generate application configuration from jinja2 template] ****************
TASK [Gathering Facts] **
[WARNING]: Host 'managed-node1' is using the discovered Python interpreter at '/usr/bin/python3.10', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.21/reference_appendices/interpreter_discovery.html for more information.
ok: [managed-node1]
TASK [Deploy rendered app conf] ***
changed: [managed-node1]
PLAY RECAP **
managed-node1 : ok=2 changed=1 unreachable=0 failed=0 skipped=0 rescued=0 ignored=0
(.venv) ubuntu@Ansible-master:~/ansible-lab$ ansible managed -m ansible.builtin.command -a "cat /opt/myapp/config/app.conf"
[WARNING]: Host 'managed-node1' is using the discovered Python interpreter at '/usr/bin/python3.10', but future installation of another Python interpreter could


ubuntu@Ansible-master: ~/ansible-lab
(.venv) ubuntu@Ansible-master:~/ansible-lab$ ansible managed -m ansible.builtin.command -a "cat /opt/myapp/config/app.conf"
[WARNING]: Host 'managed-node1' is using the discovered Python interpreter at '/usr/bin/python3.10', but future installation of another Python interpreter could cause a different interpreter to be discovered. See https://docs.ansible.com/ansible-core/2.21/reference_appendices/interpreter_discovery.html for more information.
managed-node1 | CHANGED | rc=0 >>
APPLICATION_NAME=nginx
APPLICATION_PORT=8080
APPLICATION_ENVIRONMENT=development
SERVER_HOSTNAME=managed-node1
SERVER_OS=Ubuntu
DEBUG_MODE=true
APPLICATION_DIRECTORIES:
- /opt/myapp/config
- /opt/myapp/logs
- /opt/myapp/data
(.venv) ubuntu@Ansible-master:~/ansible-lab$ nano templates/app_config.j2
(.venv) ubuntu@Ansible-master:~/ansible-lab$ ansible-playbook template_playbook.yml


ubuntu@managed-node1: ~
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
New release '24.04.5 LTS' available.
Run 'do-release-upgrade' to upgrade to it.
Last login: Tue Oct 20 12:20:53 2026 from 172.31.11.248
ubuntu@managed-node1:~$ ls -ld /opt/myapp/config/app.conf
-rw-r--r-- 1 root root 233 Oct 20 13:59 /opt/myapp/config/app.conf
ubuntu@managed-node1:~$ cat /opt/myapp/config/app.conf
APPLICATION_NAME=nginx
APPLICATION_PORT=8080
APPLICATION_ENVIRONMENT=development
SERVER_HOSTNAME=managed-node1
SERVER_OS=Ubuntu
DEBUG_MODE=true
APPLICATION_DIRECTORIES:
- /opt/myapp/config
- /opt/myapp/logs
- /opt/myapp/data
ubuntu@managed-node1:~$